

Small Neural Policies are Enough: Learning-Guided Simulated Annealing for Large-Scale Vehicle Routing

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Abstract

The Capacitated Vehicle Routing Problem (CVRP) is a fundamental NP-hard optimization problem with numerous real-world applications. We propose a substantially improved Learning-Guided Simulated Annealing (LG-SA) framework that combines the robustness of classical simulated annealing (SA) with the adaptive guidance of reinforcement learning (RL). The learned policy consists of only two small multi-layer perceptrons operating on a novel node-centric solution representation whose dimensionality is independent of the problem size, enabling efficient batched processing of thousands of instances on a single GPU. The framework is trained using a stabilized Proximal Policy Optimization (PPO) algorithm with curriculum learning to expose the policy to increasingly refined solutions. We systematically optimize the framework through a large-scale seed-paired study spanning 15 design dimensions and 78 candidate configurations, covering solution representation, architecture, reward design, optimization hyperparameters, annealing, and training strategies. The resulting framework significantly improves classical simulated annealing and establishes a new state of the art among learning-based improvement methods when jointly considering solution quality and computation time. Moreover, a single model trained on 100-customer instances generalizes without retraining to instances with up to 10,000 customers with near-linear runtime scaling. Overall, these results demonstrate that carefully designed lightweight hybrid methods can match or surpass heavyweight learning-based improvement methods while scaling far more efficiently.

1 Introduction

2 Related Works

Classical methods. Since Dantzig and Ramser (1959), the CVRP has been a central problem in combinatorial optimization, and its NP-hardness makes exact methods intractable at scale (Toth and Vigo 2002). Practical solvers are heuristic: a constructive rule (Clarke and Wright 1964) produces a first solution that local search and metaheuristics refine. Simulated annealing (SA) (Kirkpatrick, Gelatt, and Vecchi 1983) is among the simplest — a random proposal accepted by the Metropolis criterion under a decreasing temperature — and

remains attractive for its $\mathcal{O}(1)$ per-step cost. The state of the art is held by heavily engineered metaheuristics (Vidal 2022; Helsgaun 2017; Accorsi and Vigo 2021), which reach near-optimal solutions but are inherently sequential, CPU-bound, and demand substantial per-instance tuning (Liu et al. 2023).

Learning to construct. Neural combinatorial optimization instead learns a solver from data (Wu et al. 2024). *Construction* methods build a solution autoregressively. Nazari et al. (2018) adapted pointer networks (Vinyals, Fortunato, and Jaitly 2015) to the CVRP, the Attention Model (Kool, van Hoof, and Welling 2019) replaced the recurrent encoder by a Transformer, and POMO (Kwon et al. 2021) exploits solution symmetries. Since a single constructive pass is rarely near-optimal, most of the reported quality now comes from test-time search stacked on the policy (Kim, Park, and Kim 2021; Hottung, Kwon, and Tierney 2021; Choo et al. 2022), while a parallel line targets large-scale generalization (Drakulic et al. 2023; Luo et al. 2023). In all cases inference requires a full Transformer pass per construction step.

Learning to improve. *Improvement* methods start from a feasible solution and learn which local-search move to apply next, formulating the search as an MDP. NeuRewriter (Chen and Tian 2019) picks a region and a rewriting rule; L2I (Lu, Zhang, and Yang 2020) selects among improvement and perturbation operators and was the first learned method to beat LKH-3 on uniform instances, at budgets of tens of minutes per instance. Wu et al. (2022) select the node pair fed to a pairwise operator with a self-attention policy; DACT (Ma et al. 2021) adds dual-aspect embeddings and a cyclic positional encoding for the circular structure of routes; NeuOpt (Ma, Cao, and Chee 2023) extends this to flexible k -opt with guided exploration of infeasible regions, reaching some of the best learned local-search results on uniform CVRP. Others learn the operators themselves rather than their arguments (Hottung and Tierney 2022; García-Torres et al. 2022; Bui and Mai 2023). The recurring pattern is a heavyweight encoder re-evaluated at every one of thousands of search steps, which dominates the runtime and caps both the step budget and the instance size that can be afforded.

Learning-guided metaheuristics. A sparser line keeps a classical metaheuristic intact and learns only one of its components. Neural SA (Correia, Worrall, and Bondesan 2023) replaces the uniform proposal distribution of SA by a small

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equivariant policy trained with PPO, leaving the Metropolis criterion and the cooling schedule untouched; it was not evaluated on constrained routing, where capacity renders most moves infeasible. LG-SA (Andretti, Cabessa, and Strozecki 2026) transposes the idea to the CVRP with a node-pair policy and action masking. We build on that framework and push it where the above literature leaves an opening: rather than enlarging the policy, we show that a single hidden layer of 32 neurons suffices once the state representation, reward, annealing parameters and training curriculum are properly designed. This preserves the per-step cost of classical SA, turns the search into a sequence of $\mathcal{O}(N)$ batched tensor operations — thousands of instances annealing in parallel on one GPU — and shifts the comparison to the joint cost–compute trade-off, where heavyweight improvement policies are least favourable.

3 CVRP

The *Capacitated Vehicle Routing Problem (CVRP)* is defined on a complete undirected graph $G = (V, E)$, where $V = \{0\} \cup \mathcal{C}$ and $E = V \times V$ denote the sets of nodes and edges, respectively. Node 0 corresponds to the central depot, and $\mathcal{C} = \{1, \dots, N\}$ represents the set of N customers.

Each edge $(i, j) \in E$ is associated with a travel cost $c_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$, the Euclidean distance between the 2D coordinates $\mathbf{p}_i, \mathbf{p}_j \in [0, 1]^2$, satisfying $c_{ij} = c_{ji}$. Each customer $i \in \mathcal{C}$ has a non-negative demand q_i , while the depot has zero demand ($q_0 = 0$). An unlimited fleet of identical vehicles, each with uniform capacity Q , is stationed at the depot.

A route $R_k = (0, v_1, v_2, \dots, v_p, 0)$ is a simple cycle starting and ending at the depot, where $\{v_1, \dots, v_p\} \subseteq \mathcal{C}$ denotes the subset of customers served by vehicle k . A *solution* $s = \{R_1, \dots, R_m\}$ consists of a set of m routes, where the number of routes m is unconstrained. A solution s is *feasible* if and only if it satisfies the following conditions:

1. Each customer is visited exactly once by a single vehicle,

$$\sum_{k=1}^m \mathbf{1}_{\{i \in R_k\}} = 1, \quad \forall i \in \mathcal{C}.$$

2. For every route R_k , the total demand of its customers does not exceed the vehicle capacity Q , i.e.,

$$\sum_{i \in R_k \setminus \{0\}} q_i \leq Q, \quad \forall k = 1, \dots, m.$$

The set of *feasible solutions* is denoted by $\mathcal{F}$.

Let $x_{ij}(s)$ be a binary indicator variable with $x_{ij}(s) = 1$ if edge (i, j) is traversed in any route R_k of s , and $x_{ij}(s) = 0$ otherwise. The objective of the CVRP is to find a feasible solution $s^* \in \mathcal{F}$ that minimizes the total travel cost $C(s)$:

$$s^* = \arg \min_{s \in \mathcal{F}} \left\{ C(s) = \sum_{i \in V} \sum_{j \in V, i < j} c_{ij} x_{ij}(s) \right\}.$$

4 Model

We introduce an improved version of the *Learning-Guided Simulated Annealing (LG-SA)* algorithm (Andretti, Cabessa,

and Strozecki 2026; Correia, Worrall, and Bondesan 2023), a hybrid framework combining Simulated Annealing (SA) with a lightweight neural policy trained with PPO. Our model is tailored to the CVRP by means of a dedicated state representation, an efficient two-stage policy, and an optimized RL training procedure. The resulting algorithm preserves the computational efficiency of classical simulated annealing while significantly improving search effectiveness.

Overview. LG-SA starts from an initial feasible solution $s_0 \in \mathcal{F}$ and follows the standard simulated annealing search procedure. At iteration t , the current solution s_t is encoded into a node-centric representation $\bar{s}_t$ (described next). Then, an action a_t , which is a pair of nodes (i, j) , is sampled from the learned policy π_θ

$$a_t = (i, j) \sim \pi_\theta(\cdot \mid \bar{s}_t)$$

and passed to a neighborhood operator H , which transforms the current solution s_t into the candidate solution

$$s'_t = H(s_t, a_t).$$

This solution is accepted as the next solution s_{t+1} according to the classical Metropolis criterion. Consequently, LG-SA learns only the proposal distribution π_θ , while the neighborhood operator, Metropolis acceptance criterion, and cooling schedule remain those of standard SA. The overall search procedure is summarized in Algorithm 1.

Algorithm 1 LG-SA

- 1: Initialize solution s_0 and temperature T_0
 - 2: **for** $t = 0, 1, \dots$ until the stopping criterion is met **do**
 - 3: Encode current solution s_t as $\bar{s}_t$
 - 4: Sample action $a_t \sim \pi_\theta(\cdot \mid \bar{s}_t)$
 - 5: Generate candidate solution $s'_t \leftarrow H(s_t, a_t)$
 - 6: Accept s'_t as s_{t+1} according to the Metropolis criterion
 - 7: Update the temperature
 - 8: **end for**
-

Solution encoding. A solution s_t consists of an ordered sequence of customer nodes separated by depot visits. At each iteration t , LG-SA computes a node-centric representation $\bar{s}_t = \{\mathbf{x}_k\}_{k \in \mathcal{C}}$ of s_t , where each $\mathbf{x}_k$ is a feature vector that encodes both static and dynamic information associated with customer i in the current solution and search state. The encoded representation $\bar{s}_t$ constitutes the input to the neural policy π_θ .

Our feature representation combines static geometric information, dynamic descriptors of the node’s position within the current solution (e.g., predecessor and successor, local detour cost, and distance to the route centroid), route capacity statistics, and search meta-features such as the current temperature and remaining search budget. Unlike conventional geometric encodings, it also includes quality-related signals that explicitly characterize the local quality of the current solution around each node.

Importantly, the feature dimensionality is independent of the instance size, making the representation applicable to CVRP instances of arbitrary size. Moreover, every node is encoded by a fixed-size feature vector, allowing the policy

to evaluate all nodes across all instances simultaneously in a batch, through a single vectorized tensor computation. The complete feature definitions are provided in Appendix A.1.

Two-stage policy. For every solution representation $\bar{s}_t = \{\mathbf{x}_k\}_{k \in \mathcal{C}}$, the learned policy defines a distribution over node pairs, from which an action $a_t = (i, j)$ is sampled:

$$a_t = (i, j) \sim \pi_\theta(\cdot \mid \bar{s}_t).$$

Directly parameterizing this joint distribution over all $\mathcal{O}(N^2)$ node pairs is computationally expensive and unnecessary. We therefore factorize the policy as

$$\pi_\theta(a_t = (i, j) \mid \bar{s}_t) = \pi_\theta^{(1)}(i \mid \bar{s}_t) \cdot \pi_\theta^{(2)}(j \mid i, \bar{s}_t),$$

where the two conditional distributions are parameterized by two small multi-layer perceptrons (MLPs). The first network f_θ receives all node features $\{\mathbf{x}_k\}_{k \in \mathcal{C}}$ in parallel, computes a score for each node independently, and samples the first node i . Conditioned on this selection, the second network g_θ receives the pair representations $\{[\mathbf{x}_i, \mathbf{x}_k]\}_{k \in \mathcal{C}}$ in parallel also, computes a score for each candidate partner, and samples the second node j . The neural networks f_θ and g_θ are illustrated in Figure 3 (Appendix A.2).

Hence, each action-sampling step requires only $\mathcal{O}(N)$ network evaluations, instead of explicitly scoring all $\mathcal{O}(N^2)$ node pairs. Since both MLPs are lightweight and process all nodes in parallel, each sampling step can be executed efficiently on a GPU.

Neighborhood operator. The action $a_t = (i, j)$ sampled from $\pi_\theta(\cdot \mid \bar{s}_t)$ is interpreted by a neighborhood operator H , which transforms the current solution s_t into the candidate solution

$$s'_t = H(s_t, a_t)$$

by applying the corresponding local-search move.

We evaluate three standard pairwise neighborhood operators: (i) *Swap*, (ii) *Insertion*, and (iii) *2-opt/2-opt**, which respectively exchange two nodes, relocate a node, and re-connect route segments, as illustrated in Figure 4 (see Appendix A). The neighborhood operator is fixed for a given model. Consequently, the policy learns a proposal distribution specialized for the selected operator.

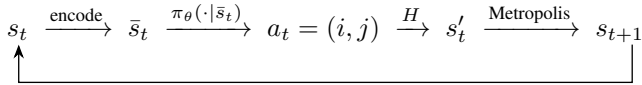


Figure 1: The LG-SA iteration loop. The current solution s_t is encoded into a node-centric representation $\bar{s}_t$, from which the learned policy $\pi_\theta(\cdot \mid \bar{s}_t)$ samples a node pair $a_t = (i, j)$. The neighborhood operator H generates a candidate solution $H(s_t, a_t)$ which is either accepted or rejected according to the Metropolis criterion.

Reinforcement learning. The LG-SA search process is formulated as a Markov Decision Process (MDP). The encoded solution $\bar{s}_t$ constitutes the state, the sampled node pair $a_t = (i, j)$ the action, and the transition combines the

neighborhood operator with the stochastic Metropolis acceptance criterion. The reward is derived from the normalized improvement in solution quality, encouraging proposals that improve the search trajectory rather than individual moves.

The model is trained using Proximal Policy Optimization (PPO) (Schulman et al. 2017). The *actor* is the two-stage policy π_θ composed of the tiny MLPs f_θ and g_θ (described above), while the *critic* is a separate small permutation-invariant MLP operating on the encoded state $\bar{s}_t$. Compared with standard PPO, our implementation incorporates an adaptive KL-regularized clipped objective, KL-driven learning-rate adaptation, and a regularized critic to improve training stability over the long stochastic SA horizon. Further details are given in Appendix A.5.

Training methodology. Beyond the PPO optimization itself, we introduce a dedicated training methodology combining diverse *initialization strategies* with *curriculum learning*.

Training instances are initialized using diverse constructive heuristics, including *random assignment*, *nearest neighbor*, *sweep*, *Clarke–Wright savings* (greedy route merging based on distance savings), and *isolate* (single-customer routes), exposing the policy to qualitatively different regions of the solution space. The construction heuristics are detailed in Appendix A.6.

During training, the reward is evaluated on short LG-SA rollouts of only a few hundred annealing steps — orders of magnitude shorter than the budgets deployed at inference. Hence, a policy trained solely from constructive solutions would rarely encounter the well-optimized states visited near the end of a long search. We therefore employ *curriculum learning* (Ma et al. 2021) to progressively shift the distribution of initial solutions encountered during training. Before each collection phase, every instance is first improved by the current policy for a limited number of LG-SA iterations, and this warm-up depth gradually increases throughout training. Consequently, early episodes focus on repairing poor initial solutions, whereas later episodes increasingly start from refined solutions and learn to perform the fine-grained improvements required near local optima. More details are given in Appendix A.7.

5 Experimental Results

5.1 Experimental Setup

Benchmark. We use three families of instances. The *design choices* of Section 5.2 are validated on data we generate ourselves following Nazari et al. (2018): coordinates uniform in $[0, 1]^2$, demands uniform in $\{1, \dots, 9\}$, and a size-dependent capacity ($Q = 20, 30, 40, 50$ for $N = 10, 20, 50, 100$); per size we pre-generate a fixed set of 10,000 instances reused by every configuration, so all comparisons are paired at the instance level. For the *baseline comparison* of Section 5.3 we instead evaluate on the test sets released by Ma, Cao, and Chee (2023), who generate instances with the same procedure and, crucially, re-run every competing method on them under a common protocol; using their data makes our numbers directly comparable to that whole table rather than to a re-derived reference. Finally, *generalization* is probed on real-world-structured and much larger instances:

CVRPLIB Set X (Uchoa et al. 2017), the XML instances of Queiroga et al. (2022), and Set XL (Queiroga et al. 2026), which reach far beyond the training size. Unless stated otherwise, results are reported at $N = 100$.

Evaluation protocol. A trained model is evaluated by running the LG-SA loop over the full test set in parallel on a single GPU, starting from random feasible solutions, with half-precision inference and a step budget stated per experiment (default 10^4). We report the mean final routing cost and the wall-clock time of the whole batch.

Statistical protocol. Every configuration is trained with 3 random seeds (5 for confirmations and final comparisons). Results are reported as the mean final cost over the test set, averaged across seeds. Comparisons are seed-paired, i.e., competing configurations are trained using the same seeds and compared through the resulting per-seed differences. Statistical significance is assessed relative to a per-batch noise floor σ_{floor} estimated from the seed-to-seed variability of stable configurations. Effects satisfying $|\Delta| > \sigma_{\text{floor}}$ are considered significant; otherwise, they are reported as ties.

5.2 Design Choices

The LG-SA design space considered in this work comprises $K = 18$ major design dimensions, each associated with a finite set of candidate configurations $\mathcal{C}_k$. These dimensions cover the neighborhood operator, feasibility strategy, node representation, MLP architectures, reward function, annealing parameters, solution initialization, optimization hyperparameters, curriculum learning, and several related design choices. Together, these dimensions define a configuration space $\mathcal{A} = \mathcal{C}_1 \times \dots \times \mathcal{C}_K$ which is far too large for exhaustive exploration. We therefore evaluate the design dimensions sequentially. At each stage, one dimension is explored while all previously selected choices are kept fixed, and the best-performing configuration is carried forward to the next stage. The final configuration is then retrained from scratch using five random seeds for confirmation.

Table 1 summarizes the outcome of our large-scale sequential design exploration: 18 design dimensions and 93 candidate configurations evaluated under the above protocol. The study identifies a set of complementary design choices that progressively refine the LG-SA design through systematic validation. The following paragraphs discuss the most influential dimensions and the rationale behind the adopted configurations. A complete description is provided in Appendices B–F.

Re-confirmed choices. Four dimensions reproduce the conclusions of Andretti, Cabessa, and Strozecki (2026) in the present setting and are not re-discussed here. Insertion with proactive masking again dominates the operator $\times$ feasibility set, with both the lowest cost and the lowest seed variance (Appendix B). Metropolis acceptance remains necessary at inference regardless of how the actor was trained — which is also why the solver keeps sampling $a_t \sim \pi_\theta(\cdot | s_t)$ rather than taking the greedy argmax, a near-deterministic proposal collapsing the diversity the acceptance step relies on (Appendix E). Random constructions remain the best training

Dimension	Adopted setting	#	Effect	App.
Operator $\times$ feasibility	insertion + masking	9	+0.29**	B
Feature set	all 13 groups	6	+0.34**	C
Pair descriptors	none	4	+0.14*	C
Critic	multi-statistic	2	+0.06**	D
Actor architecture	32×1	6	tie	D
Actor structure	indep. scorers	4	+3.20**	D
Global context	off	2	+0.51*	D
Distribution shaping	logit clip $C=10$	4	+0.17**	D
Reward	immediate	9	tie	E
Learning rates	$3 \times 10^{-4} / 10^{-3}$	9	+0.05*	F
Entropy coefficient	0.01	3	+0.13**	F
Rollout length	200	3	tie	F
PPO passes per batch	5	5	tie	F
Cooling shape	geometric	2	tie	E
Temperature range	$T_0=1, T_{\text{final}}=0.01$	6	+0.17**	E
Metropolis	on (train + test)	4	+0.12**	E
Training construction	random	5	+0.22**	E
Curriculum	ramped warm-up	10	+0.07**	G
Combined confirmation rates, rollout, T_{final}		—	+0.22**	F
<i>End-to-end effect ($N=100, 10^4$ steps, cost)</i>				
Starting point (operator winner, Table 5)			16.68 $\pm$ 0.03	
Final configuration (all winners)				TBD
Total improvement				TBD

Table 1: Master summary of the design validation, one row per dimension. *Effect*: seed-paired margin in favor of the adopted setting, against the previously adopted configuration; tie means $|\Delta| \leq \sigma_{\text{floor}}$, * that $|\Delta| > \sigma_{\text{floor}}$, ** that $|\Delta| > 2\sigma_{\text{floor}}$. # is the number of candidates evaluated. Effects come from separate sweeps and are not additive; the closing block therefore anchors the end-to-end gain on the first sweep’s winner, and is indicative rather than paired.

initialization, while the inference-time choice is washed out by 10^4 annealing steps (Appendix E). Finally, widening or deepening the two scoring networks never clears the noise floor while wall-clock rises by up to 38% (Table 12), the signature of a capacity plateau. The remaining dimensions are new to this work.

Node representation. The node representation is where this work departs most from Andretti, Cabessa, and Strozecki (2026). That descriptor was purely local — coordinates, depot flag, polar position, distance to the depot, and load ratios — whereas we add three groups that summarize a node’s *neighborhood* rather than its own position: local density ($\mu_5, \mu_{N/10}, \mu_{N/3}$, the mean distance to the nearest neighbors at three radii), the insertion detour cost, and the distance to the route centroid. Every candidate feature is constrained to be computable in closed form from the coordinate and demand tensors, so the whole 13-group state is built with batched tensor operations and no per-instance loop; a feature that cannot be vectorized this way is rejected regardless of its predictive value, since it would forfeit the throughput that makes the method competitive. The additions pay for themselves, and the geometry cluster carries most of that value. Removing the detour and centroid features — both introduced here — costs $+0.596 \pm 0.033$ in the thematic ablation (Table 9), more than

Set	Groups	Cost	Time (s)
full set (adopted)	13	16.683	173
no load ratio	12	16.648	173
set7	7	17.020	129
nodepct6	6	17.088	121
centroid6	6	17.100	122
core5	5	17.162	115

Table 2: Best-set validation at 5 seeds, seed-paired against the full set ($\sigma_{\text{floor}} \approx 0.057$). *core5* = static, meta, topology, neighbor ranks, detour; *set7* adds centroid and load share. Every pruned set is significantly worse; the only quality-neutral trim is dropping the load ratio alone. Paired deltas and per-seed views in Appendix C.

three times the gain of the best annealing or optimization setting and the joint-largest effect of that ablation, statistically tied with the relational cluster ($+0.599 \pm 0.016$); removing the density group costs a smaller but real $+0.145 \pm 0.033$. What the policy most needs is therefore not where a node sits, but how its neighborhood is shaped and what it would cost to move it.

The ablation also shows that individual group importance does not predict joint contribution. Leave-one-out and isolation tests (Appendix C) identify only 6 of the 13 groups as individually useful, yet training on those alone degrades performance (Table 2); removals compound, so density and capacity statistics carry complementary information (Table 9). The only quality-neutral trim is the route load ratio (-0.035 ± 0.106); every other pruning trades quality for speed (set7 loses 2% quality for a 25% speedup). The conditional pair descriptors of the second policy stage degrade performance ($+0.14 \pm 0.11$ and $+0.18 \pm 0.07$) at extra cost. The final model therefore keeps all 13 groups and no pair descriptors.

Stabilizing the policy. Rather than enlarging the actor, we stabilize it. Logit clipping at $C = 10$ prevents the proposal distribution from collapsing and gives the larger and more consistent gain of the two mechanisms we test, at unchanged inference cost (Table 14); a multi-statistic critic improves value estimation for free, since the critic is discarded after training (Table 11). Two further extensions fail: a pooled global instance context degrades every seed at +48% compute, and a shared encoder in place of the independent scorers loses ≈ 3 routing units in every configuration (Table 13). At $N = 100$, maintaining proposal diversity and value accuracy matters more than actor capacity.

Reward. Nine shaping variants (Table 15) show that the exact form of the dense signal is secondary — the six per-step variants fall within a band of 0.07 — but that removing intermediate credit is severe: the terminal reward loses 2.95 and the step-penalty variants 0.7 to 1.1, all with unstable training. PPO needs a dense improvement signal across the long annealing horizon.

Annealing and optimization. The cooling *shape* is inert (geometric against piecewise-constant), but its *range* pro-

duces the largest gain of any setting we actually changed (the larger margins in Table 1 are losses avoided by *keeping* an existing design, not improvements): every configuration with $T_{\text{final}} = 0.01$ beats every configuration with $T_{\text{final}} = 0.1$, at every starting temperature, while the starting temperature itself does not matter (Table 16). The low-temperature phase is exactly where the learned proposal does its fine-grained refinement, and a high final temperature truncates it. Among the PPO parameters (Tables 19 and 20), the actor learning rate dominates (main effect ≈ 0.05 against a 0.031 floor), the entropy coefficient is U-shaped with an optimum at 0.01, and rollout length is asymmetric: halving it to 50 hurts clearly, doubling it to 200 helps only marginally. Retraining all selected modifications jointly adds a further -0.223 ± 0.016 across seeds (Table 22). The number of PPO passes taken over each collected batch is the one knob where quality and stability disagree (Table 21): the mean cost falls monotonically from 1 to 10 passes, but the seed spread grows with it, the 5-to-10 gain is inside that batch’s 0.051 floor and sign-split, and at 15 passes training diverges outright on 2 of 4 seeds. Since the number of passes leaves inference cost unchanged, we adopt the tightest competitive setting, 5.

Curriculum. Training rewards are evaluated on LG-SA rollouts of only a few hundred steps, far shorter than the search budgets used at inference, so a policy trained only from raw constructive solutions rarely visits the fine-grained regime near local optima that dominates the end of a long search. Curriculum learning (Ma et al. 2021) closes this gap by moving the *starting point*: before each collection phase, the batch of instances is first warmed up by the current policy for t_{init} steps, and t_{init} ramps over training from 0 to a maximum warm-up strength ξ_{CL} over E epochs (Figure 5, Appendix A.7) — early training repairs poor solutions, late training refines already-improved ones. Two alternative mechanisms place the same idea elsewhere: a per-instance random depth $d_i \sim \mathcal{U}\{0, \dots, t_{\text{init}}\}$, which makes a single batch cover the whole spectrum from raw to deeply improved solutions, and *persistent chains*, where instances and their best solutions survive across epochs and the search resumes from them (a fraction of chains being refreshed with new instances each epoch), obtaining deep starting points at no extra warm-up compute. The three mechanisms are validated against a common 5-seed no-curriculum baseline that also provides this batch’s noise floor, followed by a budget-matched coverage ablation and a 5-seed confirmation of the winner.

That baseline is unusually reproducible (16.238 ± 0.006 over 5 seeds, $\sigma_{\text{floor}} = 0.030$), so a few hundredths of a routing unit is already a measurable effect. The ramped warm-up clears the floor by a factor of 2.4 (-0.073 ± 0.043 , improving 4 seeds out of 5) and is the setting we adopt. Neither alternative displaces it: persistent chains are uniformly harmful across their nine runs ($+0.25$ to $+0.52$, no seed ever improving) even though they cost nothing, and the random depth is a real but smaller win over the baseline (-0.034 , unanimous over 5 seeds) that loses the head-to-head against the ramp on 4 of 5 seeds. Both alternatives are monotone in the same underlying quantity — the deeper the pre-optimization, the

worse the policy — and the budget-matched ablation locates the effect precisely: at equal loop length, drawing depths from $\mathcal{U}\{0, \dots, \xi_{\text{CL}}\}$ beats warming every instance to ξ_{CL} by 0.261, but against a *constant* warm-up of the same mean depth the mix is a tie (−0.016, below the floor). What the curriculum buys is therefore a moderate warm-up rather than depth diversity per se: the actor learns from states it can still improve, and advancing the trajectory much further starves it of the reward signal it trains on (Appendix G).

5.3 Comparison with Baselines

Final model. All component winners of Table 1, the best curriculum, and the tuned optimization budget are combined into the final model, trained with 5 seeds; the best checkpoint provides the headline numbers. We compare it against blind SA under matched conditions (same operator, budget, and schedule), and we report the mean cost as a function of the SA step budget from 10^2 to 10^6 steps to characterize the anytime behavior of the guided search.

Literature baselines. Table 3 places LG-SA in the context of the published results on the Nazari benchmark. We group the baselines into four families. *Conventional solvers* — HGS (Vidal 2022) and LKH-3 (Helsgaun 2017) — delimit the quality/time frontier reachable without learning. HGS, a hybrid genetic search combining a diversity-managed population with an aggressive local search over the SWAP* neighborhood, is the stronger of the two and is the reference metaheuristic for the CVRP; we therefore report all gaps with respect to it, following Ma, Cao, and Chee (2023). The general-purpose solver OR-Tools (Furnon and Perron 2024), a common reference in earlier comparisons, is not re-run by that source and is omitted here; it is dominated by both HGS and LKH-3 on this distribution. *Construction methods* — AM (Kool, van Hoof, and Welling 2019) with collaborative policies (Kim, Park, and Kim 2021), POMO (Kwon et al. 2021), and its search-augmented variants EAS (Hottung, Kwon, and Tierney 2021) and SGBS (Choo et al. 2022) — build a solution autoregressively and obtain most of their quality from test-time sampling, beam search, or instance augmentation. *Improvement methods* — NLNS (Hottung and Tierney 2022), NCE (Kim, Park, and Kim 2021), LIH (Wu et al. 2022), DACT (Ma et al. 2021), and NeuOpt (Ma, Cao, and Chee 2023) — iteratively refine a solution and are the family LG-SA belongs to; for these we report the step budget T , which controls the quality/time trade-off. Finally, *hybrid methods* — DPDP (Kool et al. 2022) and CVAE-Opt (Hottung, Bhandari, and Tierney 2021) — couple a learned component with dynamic programming or an evolutionary search.

Reading the comparison. We source the baseline values from Ma, Cao, and Chee (2023) rather than from the older aggregation of Ma et al. (2021), because the former re-runs every baseline from its public pre-trained model on a single common test set and common hardware, instead of collecting heterogeneous numbers from the original papers. This makes the objective and time columns internally consistent, and it supplies HGS at all three sizes. Two caveats remain. First, rows marked ‡ could not be re-run and are carried over

from their original papers; their values are approximate and their run times were measured elsewhere. Second, run times still mix one GPU for the neural methods with one CPU for HGS and LKH-3, and they are totals over the whole 10,000-instance set, so the time column separates orders of magnitude rather than factors of two. The prior LG-SA rows (Andretti, Cabessa, and Strozecki 2026) carry the additional caveat of a different instance draw, marked †. Against this backdrop, the comparison we control exactly is LG-SA against blind SA under matched conditions — same operator, step budget, cooling schedule, and instances — which is reported in the same block.

5.4 Generalization and scalability.

Finally, we probe how far the node-centric representation carries across problem scales: models trained at each size are cross-tested at every other size, and the final model is evaluated on instances well beyond its training size with the same checkpoint, reporting both solution quality and runtime scaling.

6 Conclusion

Family	Method	$N = 20$			$N = 50$			$N = 100$		
		Obj.	Gap	Time	Obj.	Gap	Time	Obj.	Gap	Time
Conventional	HGS (reference)	6.130	—	10.7h	10.366	—	1.2d	15.563	—	2.5d
	LKH-3	6.135	0.08%	17.9h	10.375	0.09%	2.8d	15.647	0.54%	5.7d
Construction	AM+LCP [‡]	6.15	0.33%	23m	10.52	1.48%	52m	16.00	2.81%	2.1h
	POMO ($A=8$)	6.136	0.09%	11m	10.397	0.30%	1.4h	15.672	0.70%	7.2h
	POMO+EAS	6.132	0.04%	38m	10.379	0.13%	3.1h	15.610	0.30%	16h
	POMO+EAS+SGBS	—	—	—	—	—	—	15.579	0.10%	4.1d
Improvement	NLNS ($T=5k$)	6.175	0.73%	48m	10.506	1.35%	1.4h	15.915	2.26%	2.4h
	NCE [‡]	6.13	0.00%	11h	10.41	0.42%	2.3d	15.81	1.59%	10.4d
	LIH ($T=5k$)	—	—	—	10.544	1.72%	4.2h	16.165	3.87%	5h
	DACT ($A=6, T=10k$)	6.130	0.01%	4h	10.383	0.16%	16h	15.736	1.11%	1.7d
	NeuOpt ($D=1, T=10k$)	6.130	0.00%	41m	10.375	0.08%	2h	15.656	0.60%	4.6h
	NeuOpt ($D=5, T=40k$)	6.130	0.00%	13.7h	10.367	0.01%	1.6d	15.579	0.10%	3.8d
Hybrid	CVAE-Opt-DE [‡]	6.14	—	2.4d	10.40	—	4.7d	15.75	—	11d
	DPDP (1000k)	—	—	—	—	—	—	15.627	0.41%	1.2d
Prior LG-SA	LG-SA ($T=10^4$)	—	—	—	11.02	6.31% [†]	87s	17.36	11.55% [†]	3m
	LG-SA ($T=10^5$)	—	—	—	10.70	3.22% [†]	14m	16.80	7.95% [†]	29m
	LG-SA ($T=10^6$)	—	—	—	10.54	1.68% [†]	2.3h	—	—	—
Ours	Blind SA ($T=10^4$)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	LG-SA ($T=10^4$)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	LG-SA ($T=10^6$)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 3: Comparison with published baselines on the Nazari CVRP benchmark. Unless marked otherwise, values are taken from Ma, Cao, and Chee (2023), who re-run every baseline from its public pre-trained model on a common test set of 10,000 instances and common hardware (one GPU for neural methods, one CPU for conventional ones); gaps are relative to HGS, the strongest conventional solver. [‡]Carried over from the original paper rather than re-run, so their objectives are rounded to two decimals and their run times were measured on different hardware. [†]Prior LG-SA values are from Andretti, Cabessa, and Strozecki (2026), measured on a different draw of the same distribution; their gaps are recomputed against the HGS values above and are therefore indicative only. A and D denote the number of instance augmentations, T the number of improvement steps.

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Appendix Overview

The appendices mirror the organization of the main text. Appendix A expands the Model section component by component: the node feature definitions, the two-stage neural policy, the neighborhood operators, the feasibility strategies, the reinforcement-learning formulation with our PPO modifications, the constructive initialization heuristics, and the curriculum mechanisms. Appendices B–G then carry the evidence behind the Design Choices subsection, one appendix per design dimension in sweep order: operators and feasibility (B), node features (C), architecture (D), the search-loop components — reward, annealing, acceptance, initialization — (E), the optimization hyperparameters, including the per-seed record and mechanism of the combined-configuration divergence (F), and the curriculum study (G). Each full-results appendix collects the per-dimension motivation and candidate definitions, the complete configuration comparisons and secondary tables with paired deltas and per-batch noise floors, and the per-seed diagnostic plots behind the master summary (Table 1) and the two headline tables, and opens with the mapping from the internal configuration names appearing in the plot legends to the paper terminology. Two conventions are shared throughout: every comparison is seed-paired (both arms are trained on the same seeds), and every delta plot shows the per-seed differences against a shaded band marking that batch’s noise floor — an effect is trusted only if it clears the band with a consistent sign across seeds.

A Model Details

This appendix expands the Model section of the main text, component by component and in the same order: the node feature representation (A.1), the two-stage neural policy (A.2), the neighborhood operators (A.3), the feasibility strategies (A.4), the reinforcement-learning formulation and our PPO modifications (A.5), the constructive initialization heuristics (A.6), and the curriculum mechanisms (A.7).

A.1 Node Features

This section defines every component of the node feature vector $\mathbf{x}_i$ used by the policy and the critic. For a given solution s_t , $\text{prev}(i)$ and $\text{succ}(i)$ denote the nodes immediately before and after i in the current visiting order, $r(i)$ the route currently serving i , $Q_{r(i)}$ the total load of that route, and $d(\cdot, \cdot)$ the Euclidean distance. Figure 2 illustrates the geometric quantities, and Table 4 summarizes the groups and their dimensions. All features are normalized to comparable ranges, and each is computed locally at node i : the encoding contains no fixed-size global descriptor, which is what makes the representation applicable to any instance size.

Static instance features (6). The 2D coordinates $\mathbf{p}_i = (p_i^x, p_i^y) \in [0, 1]^2$; a depot indicator $\mathbf{1}_{\{i=0\}}$; the polar angle θ_i of the node around the depot; the distance to the depot d_{i0} , min–max normalized per instance; and the demand normalized by the vehicle capacity, q_i/Q . These features depend only on the instance, not on the current solution.

Route topology (4). The coordinates $\mathbf{p}_{\text{prev}(i)}$ and $\mathbf{p}_{\text{succ}(i)}$ of the node’s current neighbors in the visiting order. They

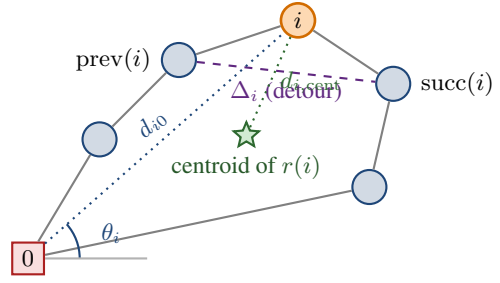


Figure 2: Geometric features attached to a node i (orange) within its current route. The detour Δ_i measures the extra length caused by visiting i between its neighbors; (θ_i, d_{i0}) locate i in polar coordinates around the depot; $d_{i,\text{cent}}$ measures how far i lies from the center of gravity of its own route.

give the policy direct access to the local shape of the route around each node.

Neighbor distance-ranks (2). For each of $\text{prev}(i)$ and $\text{succ}(i)$, the rank of that neighbor in the list of all nodes sorted by distance from i , normalized to $[0, 1]$ (the nearest node has rank ≈ 0). A well-placed node is linked to low-rank neighbors; a high rank signals a locally poor connection, independently of the absolute scale of the instance.

Relative demand (1). The demand normalized by the largest demand in the instance, $q_i / \max_{j \in C} q_j$, complementing q_i/Q with a signal that is invariant to the capacity level.

Spatial density (3). The mean distance from i to its k nearest neighbors, evaluated at three scales: $k = 5$, $k = \lceil N/10 \rceil$, and $k = \lceil N/3 \rceil$. These static statistics distinguish nodes in dense clusters from isolated ones.

Local solution quality (2). The *detour cost*

$$\Delta_i = d(\text{prev}(i), i) + d(i, \text{succ}(i)) - d(\text{prev}(i), \text{succ}(i)),$$

i.e., the extra length the tour pays for visiting i at its current position (equivalently, the saving obtained by removing it); and the distance $d_{i,\text{cent}}$ from i to the centroid of the nodes of its route, which flags spatial outliers assigned to the wrong route.

Capacity status (3). The load ratio of the node’s route, $L_{r(i)} = Q_{r(i)}/Q$; the remaining slack, $S_{r(i)} = (Q - Q_{r(i)})/Q$; and the node’s share of its route load, $\eta_i = q_i/Q_{r(i)}$. These dynamic signals indicate whether a route is nearly saturated and how much a given node contributes to that saturation.

Search meta-features (2). The current SA temperature, normalized to $[0, 1]$ over the schedule range, and the fraction of the search budget remaining. They are identical across nodes of the same state and let the policy modulate its behavior along the annealing schedule (e.g., bolder restructuring at high temperature, fine repairs near the end).

Conditional pair features (stage 2, optional). In addition to the per-node vectors above, the second stage of the policy

Group	Features	Dim
Static	$\mathbf{p}_i, \mathbf{1}_{\{i=0\}}, \theta_i, d_{i0}, q_i/Q$	6
Topology	$\mathbf{p}_{\text{prev}(i)}, \mathbf{p}_{\text{succ}(i)}$	4
Neighbor ranks	rank of $\text{prev}(i), \text{succ}(i)$	2
Relative demand	$q_i / \max_j q_j$	1
Spatial density	$\mu_5, \mu_N/10, \mu_N/3$	3
Local quality	$\Delta_i, d_{i,\text{cent}}$	2
Capacity status	$L_{r(i)}, S_{r(i)}, \eta_i$	3
Meta	temperature, remaining budget	2
Total		23

Table 4: Summary of the node feature vector $\mathbf{x}_i$.

can receive descriptors of the candidate move itself, computed only for the selected first node u and every candidate partner v : the exact insertion cost $d(v, u) + d(u, \text{succ}(v)) - d(v, \text{succ}(v))$ together with the net cost change of the full move (insertion cost minus the removal saving of u), and the normalized distance-ranks of the edges the move would create. Because they are evaluated for a single u , these descriptors preserve the $\mathcal{O}(N)$ per-step complexity. These optional descriptors are evaluated experimentally in Appendix C and are not part of the final model.

A.2 Two-Stage Neural Policy

The two-stage neural policy π_θ , parametrized by the neural networks f_θ and g_θ , is illustrated in Figure 3. Both networks are small multilayer perceptrons applied to every node (stage 1) or every candidate pair (stage 2) in parallel, as a single batched tensor operation: a linear input layer projecting to the embedding width, a stack of hidden layers of that width with LeakyReLU activations, and a bias-free linear output layer producing one scalar score per node. The adopted configuration uses an embedding width of 32 with a single hidden layer; the architecture study of Appendix D shows that larger networks bring no significant gain. The input of f_θ is the node feature vector $\mathbf{x}_k$ of Appendix A.1; the input of g_θ concatenates the feature vector of the selected node i with that of each candidate partner, dropping the duplicated search meta-features, which are identical for all nodes of a given state. The output layers are initialized orthogonally with a small gain, so the initial proposal distribution is near-uniform and early training explores like blind SA rather than following an arbitrary initial policy. Following Kool, van Hoof, and Welling (2019), the scores of both stages are bounded by $C \tanh(\cdot)$ with $C = 10$ before the softmax, which prevents the proposal distribution from collapsing to a near-deterministic choice (validated in Appendix D).

A.3 Neighborhood Operators

We consider the following standard neighborhood operators:

- *Swap*: exchanges the positions of nodes i and j , either within the same route or across two routes;
- *Insertion*: relocates node i immediately after node j ;
- *2-opt / 2-opt**: for intra-route pairs, reverses the segment between i and j ; for inter-route pairs, exchanges the route suffixes following i and j .

Figure 4 illustrates the three pairwise operators on two route fragments. The same action $a = (i, j)$ triggers a qualitatively different transformation under each operator, which is why a policy is trained for one fixed operator and learns a proposal distribution tailored to it.

A.4 Feasibility Strategies

Under capacity constraints, many actions produce an infeasible neighbor $H(s, a) \notin \mathcal{F}$ — the central obstacle in extending neural SA from unconstrained problems to the CVRP. We compare three ways of keeping the search inside $\mathcal{F}$:

- *Proactive filtering (action masking)*. The policy is restricted to the feasible subset $\mathcal{A}_f(s) = \{a : H(s, a) \in \mathcal{F}\}$ by masking infeasible actions before sampling. The conditional factorization makes this cheap: the first node i is selected freely, and only then are the feasible partners $\{j : H(s, (i, j)) \in \mathcal{F}\}$ computed and used to mask the second stage — $\mathcal{O}(N)$ checks per step instead of $\mathcal{O}(N^2)$. Every proposal is feasible by construction; since the fleet is unlimited, a node can always open a new route, so the feasible set is never empty.
- *Reactive rejection*. The policy samples any action from its learned distribution and the resulting neighbor is checked afterwards: if $s' \notin \mathcal{F}$, the move is rejected and the current solution retained. This avoids the filtering overhead but can waste both compute and learning signal when many sampled moves are infeasible.
- *Indirect representation*. Feasibility is delegated to the representation: the solution is encoded as an unconstrained permutation of the customers, the policy modifies the permutation, and a deterministic *greedy split* heuristic packs it back into capacity-feasible routes (Vidal 2016). Every state is feasible by construction and the policy never handles constraints, but the reconstruction runs at every step and its cost grows with the problem size.

A.5 Reinforcement Learning and PPO

The model is trained with Reinforcement Learning (RL) using *Proximal Policy Optimization* (PPO), the standard on-policy actor-critic algorithm (Schulman et al. 2017). To this end, we frame the LG-SA loop as a Markov Decision Process (MDP). At step t , the *state* is the encoded representation of the feasible solution $\bar{s}_t$. The *action* is the node pair $a_t = (i, j)$ handed to the operator H . The *transition* is the composition of the operator and the stochastic Metropolis acceptance: the next state is $H(s_t, a_t)$ if the move is accepted and s_t otherwise, so the environment dynamics include the acceptance randomness of SA.

The *reward* is derived from the evolution of the solution cost along the search: a move that is accepted and improves the solution receives a positive reward proportional to the (normalized) cost reduction, a rejected move yields no improvement signal, and proposals leading to infeasible neighbors can be penalized. Reward variants that instead emphasize improvements of the best solution found so far, or the final solution quality at the end of the horizon, fit the same framework; the precise shaping is a design choice studied experimentally. The agent thus maximizes the expected

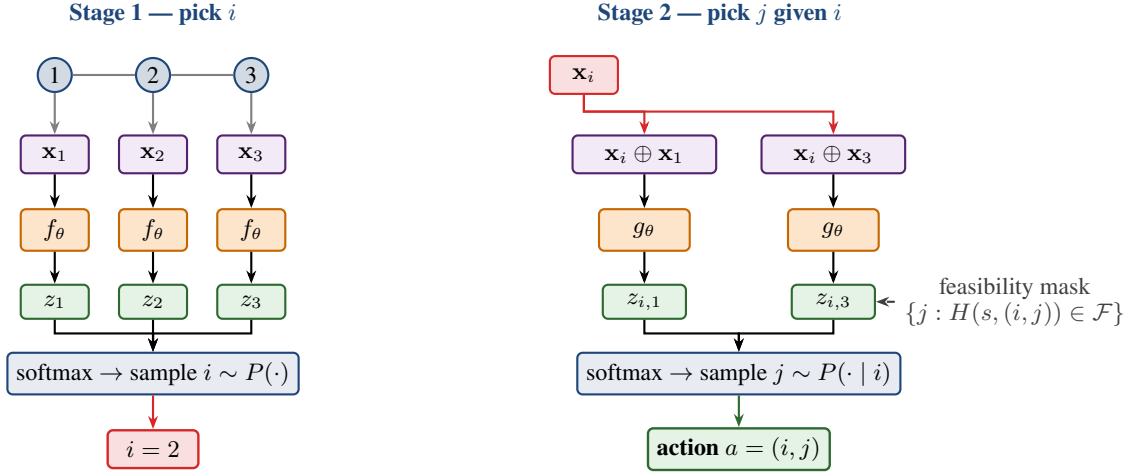


Figure 3: Two-stage conditional sampling of the node pair (i, j) , shown on a toy route of three customers. Stage 1: every node’s feature vector is scored independently by a shared network f_θ and the first node i is sampled from the softmax over the scores. Stage 2: the representation of the selected node i is combined with that of every candidate partner j and scored by a second network g_θ ; infeasible partners are masked out before the softmax, and j is sampled from the conditional distribution. The factorization $P(i, j) = P(i) \cdot P(j | i)$ makes each proposal $\mathcal{O}(N)$.

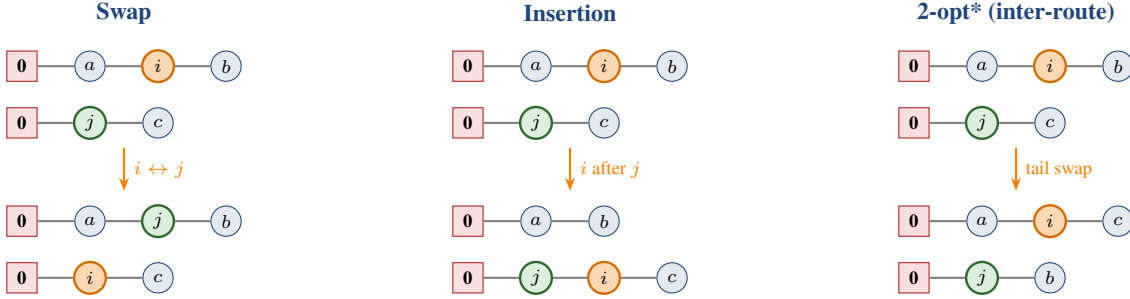


Figure 4: The three pairwise operators on two route fragments (top: before, bottom: after). *Swap* exchanges i (orange) and j (green). *Insertion* removes i and reinserts it immediately after j . *2-opt** exchanges the route suffixes following i and j ; when i and j belong to the same route, the operator instead reverses the path segment between them (standard 2-opt).

cumulative discounted reward over the SA horizon, which aligns the learned proposal with the overall goal of driving the annealing process toward low-cost solutions rather than with any single greedy step.

The policy is trained with PPO. Training alternates between a *collection* phase, in which the current policy runs the full LG-SA loop of Algorithm 1 on a large batch of freshly generated instances in parallel and stores every transition (s_t, a_t, r_t) , and an *update* phase, in which the stored episodes are replayed for several optimization epochs. Advantages are estimated with Generalized Advantage Estimation, and the actor maximizes the usual clipped surrogate objective with an entropy bonus; we refer to Schulman et al. (2017) for the standard formulation and only detail our departures from it.

The critic V_ϕ is a separate permutation-invariant network: it encodes each node feature vector independently, pools the embeddings across nodes into a fixed-size summary, and maps this summary to a scalar value estimate. Keeping the actor and critic distinct prevents interference between the

policy and value representations.

Our implementation departs from standard PPO in three ways, all aimed at stabilizing training on the long and noisy SA horizon:

- *Hybrid clipping–KL objective*. Rather than choosing between the clipped surrogate and the KL-penalized objective, we use both: an adaptive Kullback–Leibler penalty $\beta_{\text{KL}} D_{\text{KL}}(\pi_{\theta_{\text{old}}} || \pi_\theta)$ is added to the clipped loss, and its coefficient is doubled or halved after each update epoch depending on whether the measured divergence leaves a band around a target value. This soft trust region allows many optimization epochs on each collected batch without catastrophic policy drift.
- *KL-coupled learning rate*. The same divergence signal modulates the actor’s learning rate within fixed bounds — gently decreased when the policy moves too fast, increased when it stalls — so the effective step size tracks the trust region automatically.
- *Regularized critic*. The value loss is clipped, mirroring

the actor’s trust region, and augmented with a gradient penalty that encourages a smooth (near-Lipschitz) value landscape; we found this useful given that consecutive states differ by a single local move whose acceptance is itself stochastic.

A.6 Initialization Strategies

Every training episode and every evaluation starts from a feasible solution produced by one of five constructive heuristics; the parenthesized numbers give their mean construction cost on the $N = 100$ test set, spanning a sixfold range:

- *Random assignment* (57.9): the customers are placed in a uniformly random order and the sequence is split greedily into routes, a depot return being inserted whenever the next customer would exceed the remaining capacity.
- *Nearest neighbor* (19.0): routes are built one at a time by repeatedly visiting the closest unvisited customer, returning to the depot when no remaining customer fits the residual capacity.
- *Sweep* (30.5): the customers are sorted by polar angle around the depot and inserted in angular order, a new route being opened whenever the capacity would be exceeded.
- *Clarke–Wright savings* (16.4): starting from one dedicated route per customer, pairs of routes are iteratively merged in decreasing order of the distance saving achieved by the merge, subject to capacity feasibility.
- *Single-customer routes* (104.0): each customer is served by its own dedicated route — the simplest feasible construction, and by far the most expensive one.

The training-time construction fixes the state distribution the policy learns from, while the test-time construction is a free knob of the solver; the two roles are studied separately in Appendix E.

A.7 Curriculum Learning

Training episodes are much shorter than the search budgets used at inference, so a policy trained only from raw constructive solutions rarely visits the fine-grained regime near local optima that dominates the end of a long search. The curriculum (Ma et al. 2021) closes this gap by moving the *starting point*: before each collection phase, the batch of instances is first warmed up by the current policy for t_{init} steps, and t_{init} grows over training from 0 to a maximum warm-up strength ξ_{CL} over E epochs, following a linear or sigmoid ramp (Figure 5). Early in training the policy thus learns to repair poor solutions; later it increasingly learns to improve already-refined ones. A second mechanism draws a per-instance depth $d_i \sim \mathcal{U}\{0, \dots, t_{\text{init}}\}$ instead of warming every instance to the same depth, so that a single batch covers the entire spectrum from raw initializations to deeply improved solutions; it is implemented by freezing an instance once it passes d_i , so an arm of nominal strength ξ_{CL} costs the same wall-clock whether its depth is drawn or constant, and only the *effective* mean depth $\xi_{\text{CL}}/2$ differs. A third mechanism, *persistent chains*, removes the warm-up phase altogether: instances and their incumbent best solutions survive from one epoch to the next and the search resumes

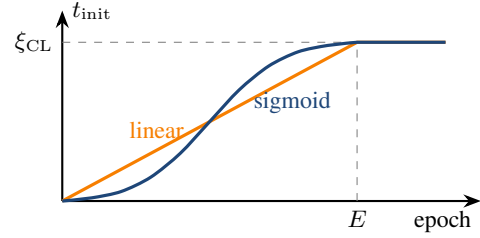


Figure 5: Curriculum ramp. Before each experience-collection phase, the batch of instances is pre-improved by running the current policy for t_{init} SA steps, and training then starts from those warmed-up solutions instead of raw constructive ones. The plot shows how that warm-up depth is scheduled: t_{init} grows from 0 (train on raw solutions) to the warm-up strength ξ_{CL} (train on deeply improved ones) over the first E epochs, either linearly or along a sigmoid, and stays at ξ_{CL} afterwards. The x -axis is training epochs, not SA steps.

where it stopped, a fraction ρ of the chains being replaced by fresh instances each epoch, so the mean resume depth is $\text{OUTER_STEPS}/\rho$ SA steps and the mechanism is free.

The curriculum is validated with its own sequential protocol: (0) a 5-seed no-curriculum baseline, which also provides the noise floor used throughout the curriculum study; (1) the ramped warm-up, at the strength, ramp duration and steepness selected in preliminary work, re-validated against that baseline; (2) a sweep of the warm-up strength under the stochastic depth, with the ramp disabled; (3) a sweep of the chain refresh rate ρ ; (4) a budget-matched coverage ablation opposing the stochastic depth to a constant warm-up of the same loop length and of the same mean depth; and (5) a 5-seed head-to-head confirmation between the two surviving mechanisms. Full results are given in Appendix G.

B Operator–Feasibility Study, Full Results

This appendix collects the complete results of the operator $\times$ feasibility sweep (Table 5); the operators and feasibility strategies themselves are defined in Appendices A.3 and A.4. The sweep replicates, under the present training configuration, the operator–feasibility study of Andretti, Cabessa, and Strozecki (2026), and reaches the same conclusion. It trains one policy per operator–strategy configuration (3 seeds per configuration, 27 runs). In the plot legends, the internal configuration names map to the paper terminology as `valid` = proactive masking, `free` = indirect representation (greedy split), `rm_depot` = reactive rejection.

Main effects. Table 6 averages each factor level over the other factor. Insertion and masking win marginally as well as jointly, so the winning configuration of Table 5 is not an interaction artefact: the best operator and the best feasibility strategy are individually best as well. Relocating a single node is the move that best matches a learned pointer to a promising node pair, whereas the value of a swap or a segment reversal depends more delicately on both endpoints simultaneously, making the credit assignment noisier

Operator	Masking	Indirect	Rejection
Insertion	16.68 ± 0.03	16.97 ± 0.04	26.36 ± 0.71
Swap	19.50 ± 1.54	22.82 ± 1.36	35.75 ± 3.04
2-opt(*)	18.93 ± 0.09	21.29 ± 0.39	25.84 ± 0.42

Table 5: Operator $\times$ feasibility study: mean final cost $\pm$ seed std at $N = 100$ (10^4 SA steps, 10,000 instances, 3 seeds; all runs start from mean cost 57.87). Insertion with proactive masking wins and is also the most stable configuration.

Operator	Cost	Feasibility	Cost
Insertion	20.00	Masking	18.37
2-opt(*)	22.02	Indirect	20.36
Swap	26.03	Rejection	29.31

Table 6: Main effects: mean final cost of each factor level averaged over the other factor. Insertion and masking win marginally as well as jointly.

— visible in the swap column’s large seed variance. Reactive rejection is uniformly the worst strategy for every operator: discarding infeasible proposals wastes both search budget and learning signal, and the policy must spend capacity learning the feasibility boundary instead of move quality. The indirect (greedy-split) representation is competitive in cost for insertion but pays the reconstruction at every step. The margin of the winning configuration over the runner-up (insertion + indirect, +0.29) is an order of magnitude above the insertion-row seed stds (0.03–0.04), and the runner-up shares the operator, so the operator choice stands even if the two leading feasibility strategies were tied.

Compute. Table 7 reports the evaluation wall-clock of every configuration. Quality and compute are essentially uncorrelated over the configurations — the cheapest column (rejection) is the worst on quality, and the most expensive configuration (2-opt(*) under masking) still loses to insertion — so the quality ranking of the main text is not a compute artefact.

Per-configuration diagnostics. Figures 6–8 give three complementary views of the set of configurations: the cost heatmap, the cost–compute trade-off, and the per-seed spread of the four best configurations. Seed variance grows quickly as configuration quality degrades, so the winning configuration is also the most reproducible one.

C Feature Ablation, Full Results

This appendix collects the protocol, complete plots, and secondary tables of the feature study behind the “Node representation” paragraph and Table 2, in the order of the main-text narrative: single-group diagnostics, thematic (cluster) removals, the head-to-head validation of the candidate best sets, the targeted single-group refinements, and the conditional pair descriptors. In the plot labels, the internal group names map to the paper terminology of Table 4 as `neighbor_rank` = neighbor distance-ranks,

Operator	Masking	Indirect	Rejection
Insertion	173	172	152
Swap	170	167	148
2-opt(*)	301	177	157

Table 7: Mean wall-clock time (s) to evaluate the full 10,000-instance test set for 10^4 SA steps. Rejection is cheapest because infeasible proposals are simply discarded; 2-opt(*) under masking pays $1.7\times$ the compute of every other configuration for its suffix-exchange feasibility checks.

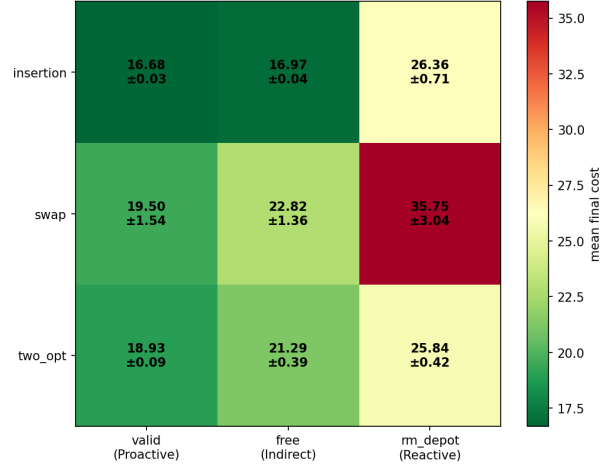


Figure 6: Mean final cost over the operator $\times$ feasibility set of configurations (green = better). The insertion/masking corner is the optimum; the rejection column is uniformly poor across all operators.

`demand_max` = relative demand, `density5/10%/33%` = $\mu_5/\mu_{N/10}/\mu_{N/3}$, `route_pct` = load ratio $L_{r(i)}$, `slack` = slack $S_{r(i)}$, and `node_pct` = load share η_i .

Protocol and reference points. The node representation is validated with three complementary protocols, all trained on the winning operator–feasibility pairing. *Leave-one-out* removes a single group from the full set and measures the degradation; *thematic* ablations remove a whole correlated cluster at once (e.g., all three density scales, or all capacity signals), exposing groups whose members are individually redundant but collectively useful; *isolation* adds a single group to a minimal base containing only the static and meta features, measuring its standalone value. From these measurements, candidate minimal feature sets are constructed — keeping groups whose removal hurts, dropping those that fail both the leave-one-out and the isolation test, and keeping one representative per correlated cluster — and confirmed seed-paired against the full set at 5 seeds; the conditional pair descriptors of the second stage are evaluated last, on top of the chosen set. The full 13-group representation reaches 16.678 ± 0.026 , against 20.084 ± 0.049 for the static+meta base: the solution-dependent features cut cost by 3.41 routing units ($\approx 17\%$). The pooled 3-seed floor of the diagnostic sweep is $\sigma \approx 0.18$, inflated by the single high-variance no-

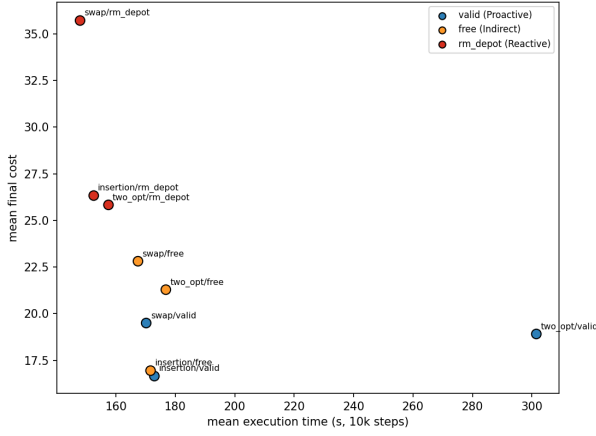


Figure 7: Cost against evaluation time for the nine configurations. Insertion/masking sits at the bottom left (best on both axes); 2-opt(*)/masking spends $1.7\times$ the compute and remains worse.

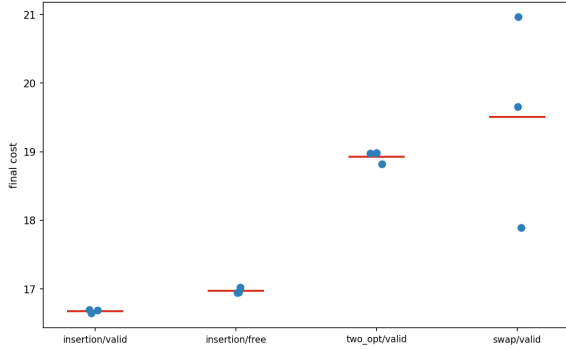


Figure 8: Per-seed final cost for the four best configurations. The winning configuration is tightly clustered; seed variance grows quickly as quality drops.

static configuration (typical seed std ≈ 0.05).

Single-group diagnostics. Table 8 reports the leave-one-out and isolation measurements. Four groups clear the noise floor at the margin: static (whose removal is catastrophic, $+3.84$), detour, topology, and meta. In isolation, the geometric and relational signals carry almost all the standalone value — detour ($+2.60$), topology ($+2.19$), neighbor ranks ($+1.26$), centroid ($+0.48$) — while the three capacity signals *hurt* when added alone to the base, and no density scale helps on its own. Applied mechanically, the single-feature pruning rule would therefore retain only six groups (static, meta, topology, neighbor ranks, detour, centroid) and drop the density, capacity, and relative-demand groups — the rule that the best-set validation below overturns. Figure 9 ranks all 30 ablation configurations and makes the two-cluster structure visible: leave-one-out runs stay near the full-set reference while isolation runs stay near the base, i.e., no single group is responsible for the 17% gap between the two. Figure 10 shows the per-seed spread of the reference configurations and

Group	Δ_{LOO}	Δ_{iso}
Static	$+3.84 \pm 0.85$	(in base)
Detour Δ_i	$+0.60 \pm 0.08$	$+2.60 \pm 0.07$
Topology	$+0.46 \pm 0.08$	$+2.19 \pm 0.15$
Meta	$+0.31 \pm 0.08$	(in base)
Slack $S_{r(i)}$	$+0.12 \pm 0.04$	-0.42 ± 0.31
Centroid $d_{i,\text{cent}}$	$+0.10 \pm 0.08$	$+0.48 \pm 0.06$
Density $\mu_{N/3}$	$+0.09 \pm 0.12$	-0.08 ± 0.06
Neighbor ranks	$+0.03 \pm 0.05$	$+1.26 \pm 0.09$
Relative demand	$+0.03 \pm 0.09$	-0.01 ± 0.02
Density μ_5	$+0.01 \pm 0.01$	-0.02 ± 0.09
Load share η_i	$+0.01 \pm 0.04$	-0.42 ± 0.10
Density $\mu_{N/10}$	-0.03 ± 0.07	-0.09 ± 0.09
Load ratio $L_{r(i)}$	-0.06 ± 0.08	-0.44 ± 0.36

Table 8: Single-group signals, sorted by marginal value. Δ_{LOO} : cost increase when the group is removed from the full set (positive = useful at the margin). Δ_{iso} : cost reduction when the group alone is added to the static+meta base (positive = useful standalone). Bold: exceeds the noise floor. Static and meta belong to the isolation base and have no standalone entry.

Theme removed	Cost	Δ vs. full
Relational (topology + nbr. ranks)	17.277	$+0.599 \pm 0.016$
Geometry (detour + centroid)	17.274	$+0.596 \pm 0.033$
Capacity (L, S, η)	17.021	$+0.343 \pm 0.059$
Density ($\mu_5, \mu_{N/10}, \mu_{N/3}$)	16.822	$+0.145 \pm 0.033$

Table 9: Thematic ablation (3 seeds, seed-paired vs. the full set; positive Δ = the theme helps). The relational and geometry clusters are the two most valuable; capacity clears the typical seed noise while density does not.

explains why the pooled 3-seed floor is inflated by the single high-variance no-static run.

Thematic ablation. Table 9 and Figure 11 report the removal of whole correlated clusters. This is the protocol that exposes collective value invisible to leave-one-out: no single capacity feature clears the floor on its own, but removing all three costs $+0.343$.

Best-set validation. The candidate sets of Table 2 were retrained head-to-head at 5 seeds on a batch with a clean floor ($\sigma_{\text{floor}} \approx 0.057$; per-set seed std between 0.04 and 0.07). Seed-paired against the full set, the pruned sets lose $+0.337 \pm 0.091$ (set7), $+0.405 \pm 0.077$ (nodepct6), $+0.417 \pm 0.049$ (centroid6), and $+0.479 \pm 0.075$ (core5) — every candidate significantly worse, degrading monotonically as groups are removed — while dropping the load ratio alone is a statistical tie (-0.035 ± 0.106). Figure 12 gives the per-seed view of the confirmation: the left panel shows that the 12-group and full-set clusters overlap and sit cleanly below every pruned candidate, and the right panel shows the seed-paired deltas against the noise floor — the evidence behind the reversal of the single-group pruning rule.

Set	Groups	Cost	Time (s)	Δ vs. full
full set	13	16.678	179	ref
– load ratio	12	16.619	166	-0.059 ± 0.076
– load ratio, – ranks	11	16.819	154	$+0.141 \pm 0.022$

Table 10: Targeted single-group refinements (3 seeds, seed-paired; mini-batch noise floor ≈ 0.035 ; per-set seed std between 0.02 and 0.05). Dropping the load ratio alone is a statistical tie with the full set; additionally dropping the neighbor ranks costs an incremental $+0.200 \pm 0.066$. The apparent 7% speedup of the 12-group set did not replicate at 5 seeds (Table 2) and is machine-load noise.

Targeted refinements. Table 10 and Figure 13 isolate the two borderline groups. Dropping the load ratio alone is a tie with the full set (reproduced independently at 3 and 5 seeds); additionally dropping the neighbor ranks costs an incremental $+0.200 \pm 0.066$, several times the floor — the neighbor ranks, not the load ratio, are the group that must stay.

Conditional pair descriptors. On top of the retained set, the two second-stage conditioning signals (edge distance-ranks and exact insertion cost, defined in Appendix A.1) were swept standalone and as a 2×2 cross-product (3 seeds, noise floor $\sigma \approx 0.071$). Both *hurt*: enabling the distance-rank descriptor costs $+0.14 \pm 0.11$ and the insertion-cost descriptor $+0.18 \pm 0.07$, each adding ≈ 20 s of wall-clock; the 2×2 main effects match the standalone deltas ($+0.13$ and $+0.17$) and the interaction is negligible (-0.02 , within the floor), so the effects are additive and the ‘both-on’ configuration is the worst (Figure 14). The extra signal apparently distracts more than it informs: the per-node features already encode the local geometry the descriptors summarize.

D Architecture Study, Full Results

This appendix collects the candidate definitions, complete set of configurations, and per-seed diagnostics of the architecture study behind the “Policy and learning” paragraph of the main text, one paragraph per design dimension in the order they were swept (critic, architecture, global context, shared encoder, distribution shaping), each sweep inheriting the previous winner. In the plot labels, the internal names map to the paper terminology as `deepsets` = multi-statistic critic, `ff` = mean-pooled feed-forward critic, `seq` = the reference actor with independent per-stage scorers, `shared` = the shared-encoder variant, `gc` = pooled global context, `bil` = bilinear second-stage head (0 = concatenation), and networks are written width/depth, e.g. `64/2L`.

Critic. The baseline critic scores each node with a small MLP and averages the scalar outputs into the value estimate, which makes V_ϕ a linear functional of the mean node representation. The alternative is the multi-statistic variant of the training section: nodes are encoded into embeddings, pooled with mean, max, and standard deviation, and mapped to the value by a nonlinear head — so dispersion information (clustered against scattered instances) survives the pooling.

Critic	Cost
multi-statistic	16.635 ± 0.038
feed-forward	16.693 ± 0.006

Table 11: Critic comparison (3 runs per arm; noise floor ≈ 0.027). The multi-statistic critic improves the mean final cost by 0.058, about twice the floor. The critic is used only during training, so inference cost is unchanged.

Net	Cost	Time (s)	Δ vs. 32×1
64×2	16.589 ± 0.004	238	-0.049 ± 0.039
64×1	16.591 ± 0.084	207	-0.047 ± 0.060
32×2	16.601 ± 0.008	187	-0.037 ± 0.030
32×1	16.637 ± 0.036	173	ref
16×2	16.649 ± 0.083	165	$+0.011 \pm 0.093$
16×1	16.668 ± 0.042	160	$+0.031 \pm 0.063$

Table 12: Actor architecture set of configurations (embedding width $\times$ hidden layers), sorted by mean cost; Δ is seed-paired against the 32×1 reference. No configuration clears the noise floor (≈ 0.053); the width main effect spans 0.069 against 0.019 for depth.

Table 11 compares the two. The critic only shapes the advantage estimates during training and is discarded afterwards, so the comparison is on solution quality alone.

Actor architecture. The embedding width and the number of hidden layers of the two scoring networks are swept jointly, trading representational ability against per-step inference cost. Table 12 gives the full width $\times$ depth configurations: the full spread from the smallest to the largest network is 0.079, no configuration clears the floor (≈ 0.053) against the 32×1 reference, and wall-clock rises by up to 38%; the faint monotone trend favors width over depth (main-effect spans 0.069 against 0.019). Figure 15 adds the seed-paired deltas against the noise band — every configuration falls inside it — and Figure 16 shows the cost–compute view: quality improves only marginally along a steeply rising compute axis, the signature of a capacity plateau.

Global context. The *global context* option pools the node representations into a summary g (mean, max, and standard deviation) appended to the input of both stages — the cheapest way to give the per-node scores access to instance-level information, at one pooling operation per step. It backfires: the mean cost rises from 16.633 to 17.143 (seed-paired $+0.508 \pm 0.496$, one seed degrading by $+1.08$) and the evaluation wall-clock from 180 to 255 s ($+48\%$) — the instance-level summary distracts the per-node scores rather than informing them. Figure 17 shows the per-seed effect: every seed degrades, one dramatically, so the regression is not a variance artefact.

Shared encoder. Figure 18 illustrates the variant: each node is encoded once into an embedding $\mathbf{h}_i$, and both selection stages are lightweight heads over the shared embeddings, the conditional score being computed either by a concatenation head over $[\mathbf{h}_j \parallel \mathbf{h}_i]$ or as a bilinear compatibility

Stage-2 head	Context	Cost	Time (s)	Δ vs. ref
bilinear	on	19.830	196	$+3.196 \pm 0.103$
bilinear	off	19.860	168	$+3.226 \pm 0.163$
concat	off	20.154	192	$+3.520 \pm 0.251$
concat	on	20.160	236	$+3.526 \pm 0.099$

Table 13: Shared-encoder set of configurations, sorted by mean cost; Δ is seed-paired against the reference actor re-run with context off (16.633 ± 0.036 , 180 s). Per-configuration seed std between 0.06 and 0.26; noise floor ≈ 0.176 . Within the shared family the bilinear head beats concatenation (main effect -0.31) and the context toggle is neutral (-0.01), but every configuration is ≈ 3 units worse than the reference.

Clipping	Temp.	Cost	Time (s)	Δ vs. ref
$C = 10$	fixed	16.457	177	-0.173 ± 0.059
$C = 10$	learned	16.481	179	-0.149 ± 0.085
off	learned	16.580	176	-0.050 ± 0.069
off	fixed	16.630	175	ref

Table 14: Distribution-shaping set of configurations, sorted by mean cost; Δ is seed-paired against the both-off reference. Per-configuration seed std between 0.04 and 0.11; noise floor ≈ 0.072 . Only logit clipping clears the floor; the learnable temperature does not help it (clipping main effect -0.136 , temperature -0.014).

$\mathbf{q}^\top \mathbf{k} / \sqrt{d}$ between a query built from the selected node and a key per candidate. Table 13 and Figure 19 report its 2×2 set of configurations (head type $\times$ context) against the reference actor: every configuration sits $+3.20$ to $+3.53$ above it — a 19% quality loss, more than $18\times$ that batch’s noise floor — so forcing both stages through one trunk destroys the pair-scoring ability at this problem size; the failure is structural rather than a bad corner of the set of configurations.

Distribution shaping. Two flags tame the proposal distribution without adding performance: *logit clipping*, which bounds the logits to $C \tanh(z)$ before the softmax so the distribution cannot collapse to a near-deterministic choice (Kool, van Hoof, and Welling 2019), and a *learnable softmax temperature* per stage. Table 14 and Figure 20 report their 2×2 set of configurations: only clipping clears the floor (all three seeds improve), while the learnable temperature is within noise on its own (-0.050 ± 0.069) and slightly degrades the clipped configuration when stacked, with doubled seed variance — one more trainable knob for PPO to destabilize.

E Search-Loop Components, Full Results

This appendix collects the full rankings and per-seed diagnostics behind the reward, annealing, acceptance, and initialization studies of the main text, one paragraph per design dimension. All four batches follow the standard protocol (3 seeds, 10,000 SA steps at half precision on the Nazari $N = 100$ test set); the reference configuration reproduces at 16.459–16.460 across the four batches, matching the adopted

configuration of the architecture study (Table 14).

Reward. The reward decides which behavior PPO reinforces: per-step improvement, new records, or only the final outcome. Because the SA loop already owns part of the exploration — worsening moves are accepted by the Metropolis criterion, not chosen by the policy — it is not obvious how much of the trajectory should be credited to the actor. Let Δ_t denote the cost improvement realized by the executed transition (zero when the move is rejected) and B_t the best cost found so far. The compared variants are: *immediate*, $r_t = \Delta_t$ with a fixed penalty for infeasible proposals; *acceptance-aligned*, which rewards accepted improving moves by Δ_t but deliberately assigns zero to accepted worsening moves (exploration is SA’s responsibility, not the policy’s fault) and a small constant penalty to rejected proposals; *best-so-far*, $r_t = \max(0, B_{t-1} - B_t)$, which only rewards new records; *terminal*, a single end-of-episode reward equal to the relative improvement over the initial cost; *mixtures*, either a fixed convex combination of the immediate and best-so-far signals or a scheduled interpolation that starts from the dense acceptance-aligned signal early in training and shifts toward the sparser best-so-far or terminal signals; and *step-penalty*, a negative per-step reward proportional to the current best cost, pressuring fast descent. In the plot labels, the internal reward names map to this terminology as `immediate` = immediate, `sa_aligned` = acceptance-aligned, `global_best` = best-so-far, `hybrid` = the fixed mixture ($\alpha = 0.5$), `curriculum` and `curriculum_terminal` = the scheduled interpolations toward best-so-far and terminal respectively, and `step_penalty_init` and `step_penalty_16` = the per-step penalty normalized by the initial cost and by a fixed constant close to the typical final cost.

Table 15 splits the nine candidates into two regimes (noise floor 0.042, the pooled seed std of the stable cluster). Every reward built on dense per-step improvement lands in a band only 0.07 wide: the reference immediate reward is best at 16.459, the fixed mixture is a statistical tie (seed-paired $+0.001 \pm 0.042$), and the acceptance-aligned, best-so-far, and scheduled variants trail by at most 0.07 — the actor is largely insensitive to the exact shaping as long as the signal is dense. Removing the per-step credit is catastrophic: the terminal reward loses 2.95 routing units with seed swings of almost a full unit, and the step-penalty schemes lose 0.7 to 1.1 with similarly unstable training — the penalty fights the improvement signal instead of complementing it. Figure 21 zooms on the stable cluster and shows the seed-paired deltas against the noise band.

Cooling schedule. The cooling schedule controls how quickly the acceptance criterion hardens from tolerant exploration to strict descent; a learned proposal could in principle prefer a different cooling profile than blind SA. The *shape* comparison holds the temperature range (T_0, T_{final}) fixed and opposes the *geometric* decay $T_{k+1} = \alpha T_k$ to a piecewise-constant *step* schedule that drops the temperature at one half, two thirds, and three quarters of the horizon. It is a statistical tie: 16.460 against 16.473, a seed-paired -0.013 ± 0.034 against a noise floor of 0.044, with the per-seed sign split

Reward	Cost
immediate (reference)	16.459
fixed mixture	16.460
sched. $\rightarrow$ best-so-far	16.503
best-so-far	16.510
acceptance-aligned	16.516
sched. $\rightarrow$ terminal	16.531
step-penalty (const.)	17.198
step-penalty (init)	17.549
terminal	19.411

Table 15: Reward ablation: mean final cost over 3 seeds, sorted; the rule separates the dense stable cluster from the sparse and penalty-based rewards. Wall-clock is flat across arms (173–177 s). Per-seed diagnostics in Figure 21.

$T_0 \backslash T_{\text{final}}$	0.01	0.1
0.5	16.342	16.475
1	16.343	16.508
2	16.354	16.486

Table 16: Temperature-rangeset of configurations: mean final cost over 3 seeds; rows = initial temperature T_0 , columns = final temperature T_{final} (reference in bold).

and identical wall-clock (Figure 22). The *range*, by contrast, holds the largest single-knob improvement of the whole validation: a 3×2 set of configurations retrains the policy over $T_0 \in \{0.5, 1, 2\}$ and $T_{\text{final}} \in \{0.01, 0.1\}$ (Table 16) and splits cleanly by final temperature — every $T_{\text{final}} = 0.01$ configuration beats every $T_{\text{final}} = 0.1$ configuration, and the reference range (1, 0.1) is in fact the worst configuration. Cooling deeper gains -0.165 ± 0.034 at the reference T_0 , four times the 0.041 floor, with a unanimous sign at every starting temperature; the initial temperature is inert (at $T_{\text{final}} = 0.01$ the three starting points are a three-way tie). Since 0.01 is the lowest tested value, an even lower floor cannot be ruled out.

Metropolis acceptance. Once the proposal is learned, does the stochastic acceptance still earn its keep — and should the policy be trained with it in the loop? A 2×2 ablation enables or disables the Metropolis criterion independently at training and at test time (the “off” arm accepts every proposed move), quantifying whether the learned proposal replaces the stochastic acceptance or complements it. The cross (Table 17) shows the criterion earns its keep at inference, not at training: removing the acceptance at test time costs $+0.119 \pm 0.025$ over the Metropolis-trained actor and $+0.094 \pm 0.009$ over the greedy-trained one — several times the 0.034 floor and unanimous across seeds — whereas the training-time flag is within noise at the standard test-on inference ($+0.027 \pm 0.045$, sign split). Only when deploying greedily does greedy training pay ($+0.052 \pm 0.050$), a mild train/test-consistency effect outside the regime we deploy in. Figure 23 decomposes the cross into its two effects; the greedy-trained arms also show markedly lower seed variance

Train \ Test	Metropolis	Accept-all
Metropolis (reference)	16.459	16.578
Accept-all	16.432	16.526

Table 17: Metropolis 2×2 cross: mean final cost over 3 seeds; rows = training setting, columns = inference setting. Per-seed effects in Figure 23.

(0.010/0.020 against 0.040/0.050).

Initialization. Every episode starts from a constructive solution, both during training and at inference, and the two roles need not agree: at training time the construction fixes the state distribution the policy learns from, while at inference it is a free knob of the solver. Training starts from a single constructive heuristic — random assignment, nearest neighbor, sweep, or single-customer routes — or from a progressively introduced mixture of all four; every trained policy is additionally cross-tested from all five test-time constructions (the four above plus Clarke–Wright savings). The heuristics appear under their code names *random* (random assignment), *sweep*, *nearest_neighbor*, *Clark_and_Wright* (Clarke–Wright savings), and *isolate* (single-customer routes); their construction costs on the test set are 57.9, 30.5, 19.0, 16.4, and 104.0 respectively. Table 18 gives the full train $\times$ test cross. At inference the construction barely matters — the random-trained actor lands in a band only 0.160 wide, the one real edge being a Clarke–Wright start, which already sits at the fixed point of the search (-0.153 ± 0.021 against the 0.063 stable floor). At training time it is decisive: actors trained from the structured heuristics lose 1.1 to 2.8 routing units at the standard inference, unanimously and far past even the global 0.395 floor, and mixing all four constructions into training dilutes coverage, a small but unanimous regression ($+0.222 \pm 0.084$). Figure 24 shows the cross as a heatmap, and Figure 25 shows the two slices that support the main-text reading: training construction matters, inference construction barely does, and the matched train/test diagonal — worse than the random-trained actor everywhere — rules out a mismatch explanation.

F Optimization Hyperparameters, Full Results

This appendix documents the optimization-hyperparameter sweeps and their combined-configuration validation, in particular the reproducible divergence that fixed the final critic learning rate. Every design dimension before this one compared design alternatives; the plain numerical knobs of the optimizer were inherited, not tuned. With the architecture and search loop frozen, the knobs are swept around the reference configuration one dimension at a time: the actor and critic learning rates (a 3×3 set of configurations), the entropy coefficient, and the rollout length — the number of SA steps collected per PPO update, which sets both the horizon over which advantages are estimated and how far into the cooling schedule the collected experience reaches. Every reference value lies strictly inside its configuration set, so each sweep

Train \ Test	Rand.	Sweep	NN	C-W	Isol.
random (reference)	16.460	16.467	16.467	16.307	16.444
sweep	17.548	16.950	16.854	16.271	17.148
nearest neighbor	19.228	17.729	17.401	16.381	18.837
single-customer	17.736	17.725	17.753	16.442	17.706
mixed	16.682	16.663	16.700	16.357	16.673

Table 18: Initialization cross: mean final cost over 3 seeds, rows = training construction, columns = test-time construction (NN = nearest neighbor, C-W = Clarke-Wright savings, Isol. = single-customer routes). The random-trained row is uniformly the best, and its spread across test constructions is only 0.160 although the construction costs span 16.4 (C-W) to 104.0 (single-customer). The structured-training rows recover only in the C-W column, where the construction itself already sits at the fixed point of the search — that column measures the inference, not the actor. Worst case over inferences: 16.467 (random), 16.700 (mixed), 17.5–19.2 (structured).

Actor \ Critic	3×10^{-4}	10^{-3}	3×10^{-3}
3×10^{-4}	16.447	16.435	16.396
10^{-3}	16.509	16.460	16.466
3×10^{-3}	16.509	16.502	16.510

Table 19: Learning-rate set of configurations: mean final cost over 3 seeds; rows = actor rate, columns = critic rate (reference in bold).

is a proper local search around the default. All batches follow the standard protocol (3 seeds per configuration, 10,000 SA steps at half precision on the Nazari $N = 100$ test set); the per-batch noise floors are 0.031 (learning rates), 0.046 (entropy), 0.041 (rollout), and 0.041 (temperature range), and the reference configuration reproduces at 16.459–16.508 across the four batches. A fifth sweep, the number of PPO passes per collected batch, is run at the end of the chain on the configuration that all the above produce, and therefore carries its own floor and its own reference level.

Learning rates. The actor learning rate is the dimension that matters, and its effect is monotone (Table 19): the three worst configurations all share the highest actor rate, the three best all share the lowest, and each step increase in the actor learning rate costs roughly 0.05 on average across critic-rate settings. The critic rate is a weaker, second-order knob. The best configuration — actor 3×10^{-4} , critic 3×10^{-3} — improves on the reference by -0.064 ± 0.051 , unanimous across seeds against the 0.031 floor; the combined-configuration runs below explain why the adopted configuration nevertheless keeps the critic rate at 10^{-3} .

Entropy and rollout. The entropy coefficient is a clean confirmation of the default (Table 20, top): 0.01 sits at the bottom of a U, with both neighbors unanimously worse — removing the bonus costs $+0.128 \pm 0.021$ (the policy under-explores and collapses onto a narrow move set), overweighting it costs $+0.783 \pm 0.036$ (near-uniform proposals).

Entropy coefficient	Cost
0	16.587
0.01 (reference)	16.459
0.05	17.242
Rollout length	
50	16.904
100 (reference)	16.460
200	16.404

Table 20: Entropy-coefficient and rollout-length sweeps: mean final cost over 3 seeds.

The rollout length is asymmetric (bottom): halving it to 50 is a large unanimous regression ($+0.444 \pm 0.015$) — the collected experience never reaches the cool end of the schedule where the hard moves live — while doubling it to 200 is a marginal, sign-split gain (-0.056 ± 0.053) at twice the collection cost per update.

PPO passes per batch. The number of optimization passes the actor and critic take over each collected batch was held at 3 for every component sweep, to keep training cheap, and is tuned once at the end of the chain on the no-curriculum configuration (3 seeds per arm, 4 for the last one; noise floor 0.051, pooled over the stable arms). Table 21 shows the trade-off. The mean improves monotonically from 1 to 10 passes, for a total of 0.081, which is the expected shape: more gradient steps extract more signal from the same experience. But the seed spread travels with it — the 10-pass arm is three times as wide as the 5-pass arm, and its mean rests on two good seeds against a third that regresses — and the improvement it buys, -0.022 ± 0.093 , is inside the floor and sign-split, hence a tie. One grid step further, at 15 passes, the updates are large enough to destabilize training outright: two of four seeds end at 19.57 and 20.10, roughly 23% above a converged solution, and the failure is frequent rather than anecdotal (the fourth seed was added precisely to test whether the first divergence was a one-off). Since the number of passes is a training cost only — inference is a fixed SA rollout and its wall-clock is flat across arms at 169–174 s — the decision is governed by stability and training time, and 5 dominates its only rival on both while matching it to within the noise.

Combined-configuration runs. The three wins — the lower actor rate, the longer rollout, and the deeper cooling floor of the annealing study — were each measured in isolation, so the joint configuration is retrained from scratch to test additivity. With the best critic rate of 3×10^{-3} , five of six seeds land at 16.216 ± 0.048 , matching the sum of the individual effects; reducing the critic rate to the reference 10^{-3} converges on every seed at 16.230 ± 0.039 , a seed-paired -0.223 ± 0.016 against the reference configuration, unanimous — the adopted configuration. Two caveats carry over to the main text: the two winning values (actor rate and cooling floor) both sit on the boundary of their explored ranges, and the divergence below shows that stacking individually-validated winners can create interactions none

PPO passes	Cost	Std	Δ vs. 5
1	16.343	0.056	$+0.059 \pm 0.045$
3	16.319	0.023	$+0.036 \pm 0.023$
5 (adopted)	16.284	0.024	—
10	16.262	0.078	-0.022 ± 0.093
15	18.069	2.049	$+1.286 \pm 2.203$

Table 21: PPO passes per collected batch: mean final cost over 3 seeds (4 for the last arm), with seed-paired deltas against the adopted value on the shared seeds. Noise floor 0.051, pooled over the stable arms above the rule; the 15-pass arm is excluded from the floor because its spread is divergence, not sampling noise — two of its four seeds end at 19.57 and 20.10, and its converged-seed mean (16.305) is no better than the 3-pass arm.

Seed	Critic 3×10^{-3}	Critic 10^{-3}
0		16.265
1	19.865 / 19.872	16.268
2		16.219
3		16.258
4		16.152
5		16.186
mean (converged)	16.216 ± 0.048	16.230 ± 0.039

Table 22: Per-seed final cost of the combined configuration (actor rate 3×10^{-4} , rollout 200, $T_{\text{final}} = 0.01$) under the two candidate critic learning rates. Seed 1 under the 3×10^{-3} critic was trained twice independently; both runs diverge into the same degenerate basin. The 10^{-3} critic converges on every seed.

of them exhibits alone. Table 22 lists every training run of the two combined configurations. The aggressive variant (critic rate 3×10^{-3} , the best configuration of the learning-rate sweep) was extended from 3 to 6 seeds after the first divergence, and the failing seed was retrained a second time from scratch: both runs land at 19.86–19.87, so the failure is deterministic given the seed, not a flaky run. The same seed trains without incident under the reference configuration (16.473) and under the final configuration (16.268), so the seed itself is not pathological — the divergence is created only by the combination of knobs, each of which was stable across all its seeds in isolation.

Failure mechanism. The quantity that overflows is not the learning rate itself but the product of learning rate and advantage, and it is the fast critic that manufactures the oversized advantages. With a 200-step rollout cooling to $T_{\text{final}} = 0.01$, the long low-temperature tail of every collected episode is a regime where the Metropolis rule rejects nearly every non-improving move; a rejected move leaves the state unchanged and its immediate reward is exactly zero, so early in training — when the near-random policy finds few improving moves — the return signal is mostly zeros punctuated by rare spikes. A critic stepping $10\times$ faster than the policy it evaluates overshoots when fitting that target, and an occasional

badly mis-scaled value estimate produces one outsized advantage, hence one outsized policy step; the updated policy then jumps far enough from the collection policy that the exponentiated log-ratio overflows in the backward pass and the run never recovers. The seed dependence is a race: a seed that lingers longer in the flat zero-reward regime gives the fast critic more opportunities to overshoot before the policy finds signal — the failing seed spent roughly 50 epochs there where a converging seed escaped in about 5. Matching the critic rate to the reference 10^{-3} keeps the advantages bounded and closes the race at no measurable cost: the two variants are statistically indistinguishable on their common converged seeds (Table 22).

G Curriculum Study, Full Results

This appendix collects the complete curriculum results: the no-curriculum baseline and the noise floor it defines, the three mechanisms of Appendix A.7, the budget-matched coverage ablation, and the 5-seed confirmation that settles the choice. The whole study sits on the configuration frozen by all preceding design dimensions, with the curriculum as the only free component; it therefore has its own reference level and its own noise floor, and none of its numbers may be compared across batch boundaries. In the sweep configuration files, `CL_MODE=warmup/chain` selects the warm-up or the persistent-chain mechanism, `MAX_OUTER_STEPS_CL` is the warm-up strength ξ_{CL} , `MAX_PROB_STEP` the ramp duration E , `STEEP_SIG` the sigmoid steepness κ , `CL_RANDOM_DEPTH` the per-instance stochastic depth, and `CHAIN_REFRESH` the chain refresh rate ρ . Setting `CL_TYPE=lin` together with $E = 1$ disables the ramp, which is how the stochastic-depth arms isolate ξ_{CL} as their only knob. All arms are evaluated under the standard protocol (10,000 SA steps at half precision on the Nazari $N = 100$ test set, identical instances throughout, mean initial cost 57.87).

Baseline and noise floor. The no-curriculum baseline reaches 16.238 with a seed-to-seed standard deviation of 0.006 over 5 seeds (16.2277, 16.2356, 16.2405, 16.2411, 16.2437) — five seeds spanning 0.016 routing units end to end, which is itself a check that the frozen configuration trains reproducibly. The floor used throughout this appendix is the more conservative $\sigma_{\text{floor}} = 0.030$, the pooled within-configuration standard deviation of the baseline and the ramped warm-up; it is applied unchanged to every comparison below so that all verdicts are commensurable. Arms trained on 3 seeds are paired against the 3-seed subset of the baseline (16.2348), never against its 5-seed mean.

The ramped warm-up survives, on four seeds out of five. The warm-up ramp improves the baseline by -0.073 ± 0.043 , about 2.4 times the floor, so the mechanism is emphatically not a null (Table 23). The effect is nonetheless split: four seeds improve by 0.079 to 0.106, while one moves by $+0.002$, statistically identical to its baseline value. That single non-responder is what inflates the arm’s spread (0.042 against the baseline’s 0.006) and breaks unanimity, and the same seed reappears in every split verdict of this study.

Mechanism	Cost	Std	Δ vs. none	Resp.
no curriculum	16.238	0.006	—	—
ramped warm-up	16.165	0.042	-0.073 ± 0.043	4/5
stochastic depth	16.204	0.015	-0.034 ± 0.015	3/5
persistent chains	16.487	0.053	$+0.253 \pm 0.047$	0/3

Table 23: The three curriculum mechanisms at their best setting, seed-paired against the no-curriculum baseline; 5 seeds except persistent chains (3 seeds, paired against the matching baseline subset). *Resp.* counts the seeds that individually improve past the floor $\sigma_{\text{floor}} = 0.030$. The ramped warm-up uses $\xi_{\text{CL}} = 250$, $E = 2000$, $\kappa = 0.01$; the stochastic depth $\xi_{\text{CL}} = 250$ with the ramp disabled; the chains $\rho = 0.2$.

Mechanism	Setting	Depth	Δ vs. none
stochastic depth	$\xi_{\text{CL}} = 250$	125	-0.037 ± 0.011
stochastic depth	$\xi_{\text{CL}} = 500$	250	$+0.135 \pm 0.037$
stochastic depth	$\xi_{\text{CL}} = 1000$	500	$+0.302 \pm 0.033$
persistent chains	$\rho = 0.20$	1000	$+0.253 \pm 0.047$
persistent chains	$\rho = 0.10$	2000	$+0.390 \pm 0.042$
persistent chains	$\rho = 0.05$	4000	$+0.523 \pm 0.084$

Table 24: All stochastic-depth and persistent-chain arms (3 seeds each, 18 runs), ordered by effective pre-optimization depth in SA steps ($\xi_{\text{CL}}/2$ for the warm-up, $\text{OUTER_STEPS}/\rho$ for the chains, with $\text{OUTER_STEPS} = 200$). Deltas are seed-paired against the 3-seed baseline subset; every arm is unanimous across its seeds. Only the shallowest arm improves on no curriculum at all.

Both alternatives are monotone in pre-optimization depth. The stochastic depth is strictly monotone in ξ_{CL} and only its shallowest arm improves on the baseline; the chains are monotone in ρ , i.e. they improve as the mechanism is used *less*, and their limit $\rho \rightarrow 1$ is the baseline by construction. Table 24 places all six arms on the single axis they share, the mean number of SA steps by which the trajectory has been advanced before the actor collects experience. The reading is consistent: the shallower the pre-optimization, the better the policy — the actor learns from states it can still improve, and advancing the trajectory past a few hundred steps starves it of the reward signal it trains on, compounding the long zero-reward tail that the deep cooling floor already produces. Two caveats block reading the table as one curve. The two families are disjoint in depth (125–500 against 1000–4000), so mechanism and depth are confounded; and at the boundary the ordering inverts, the deepest warm-up arm being worse than a chain arm at twice its depth. Persistent chains are therefore rejected on their own evidence — nine runs, all significantly worse than doing nothing, with the sweep’s own gradient pointing back at the baseline — rather than as a corollary of a depth law.

Coverage is a mean-depth effect, not a mixing effect. The advantage of the stochastic depth could be the depth *mix* — different instances warmed to different depths within one batch — or simply its lower mean depth. Two constant-depth

Warm-up	Depth	Cost	Δ vs. none
constant, $\xi_{\text{CL}} = 250$	250	16.459	+0.224
stochastic, $\xi_{\text{CL}} = 250$	125	16.198	−0.037
constant, $\xi_{\text{CL}} = 125$	125	16.214	−0.020

Table 25: Coverage ablation, 3 seeds per arm, seed-paired against the baseline subset (16.2348); *Depth* is the mean number of warm-up SA steps. The first row matches the stochastic arm’s loop length (and wall-clock, since freezing an instance does not shorten the loop), the third its effective depth. All deltas are unanimous except the last, which is split and below the floor.

controls separate the two (Table 25). At matched loop length, hence matched wall-clock, the mix is worth -0.261 against a constant warm-up to $\xi_{\text{CL}} = 250$, unanimously and about nine times the floor; and that constant arm is not merely worse than the mix but $+0.224$ *above the baseline*, i.e. actively harmful, exactly the over-optimization the monotonicity above predicts. The sharper control is the constant warm-up at the mix’s own effective depth, and there the edge collapses to -0.016 , below the floor and split across seeds. Mean depth is what matters; mixing per se is a wash at the mean. Its one measurable effect is local: the entire gap against the matched-depth control comes from the same hard seed that the ramp cannot move, which the mix pulls from 16.330 to 16.190 while leaving the other seeds marginally worse.

Confirmation. The two surviving mechanisms are compared head to head on all 5 seeds, the decision rule fixed in advance being to prefer the simpler stochastic depth only if it wins or ties unanimously (Table 26). It does neither: the ramp leads by 0.039 ± 0.053 on the mean, above the crude floor, and 4 of the 5 seeds prefer it — the exception being, once more, the seed the ramp fails to move. Two qualifications cut against over-reading the margin. The paired 95% confidence interval, $[-0.027, +0.105]$, includes zero, so by a strict test the two are tied; but the tie is not symmetric, since going from 3 to 5 seeds moved the point estimate from 0.018 to 0.039, i.e. further toward the ramp, the two added seeds behaving like its responders rather than like the hard seed. The ramped warm-up is adopted. The stochastic depth remains the simpler and more per-seed-uniform runner-up — two fewer hyperparameters, no ramp duration and no steepness — and the one regime in which it might still pay off, out-of-distribution evaluation, is untested here.

Seed	Ramp	Stoch. depth	Δ
0	16.237	16.190	-0.047
1	16.162	16.201	+0.039
2	16.142	16.204	+0.062
3	16.135	16.229	+0.095
4	16.148	16.196	+0.047
mean	16.165 $\pm$ 0.042	16.204 $\pm$ 0.015	+0.039 $\pm$ 0.053

Table 26: Five-seed confirmation, ramped warm-up against stochastic depth at $\xi_{\text{CL}} = 250$; Δ is stochastic - ramp, so a positive value favors the ramp. Bold marks the better configuration on each seed. Seed 0 is the recurring hard seed that the ramp does not move and that the depth mix rescues.

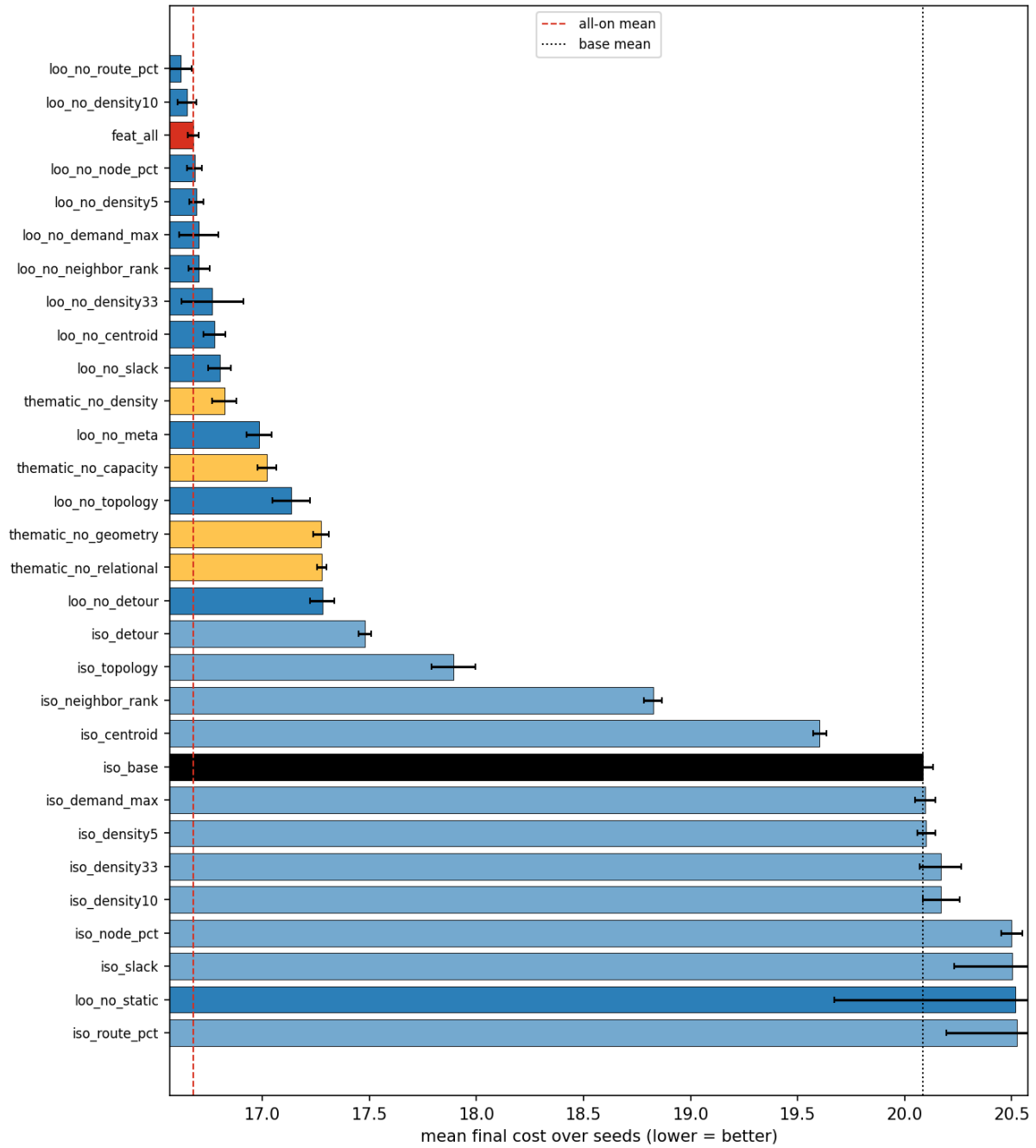


Figure 9: All 30 ablation configurations ranked by mean final cost. Leave-one-out configurations (dark blue) cluster tightly near the full-set reference (red dashed); isolation configurations (light blue) sit near the static+meta base (dotted).

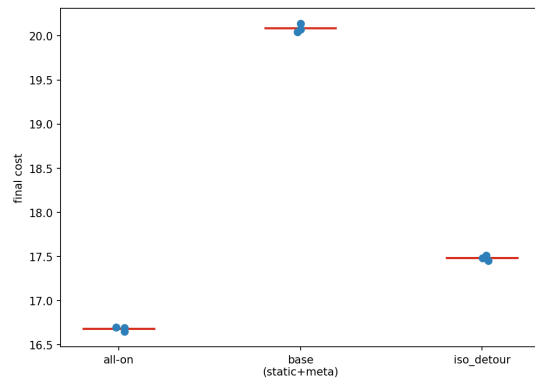


Figure 10: Per-seed final cost for the key reference configurations (red bars = means). Seed variance is tiny except for the no-static configuration, which inflates the pooled noise floor of the 3-seed sweep.

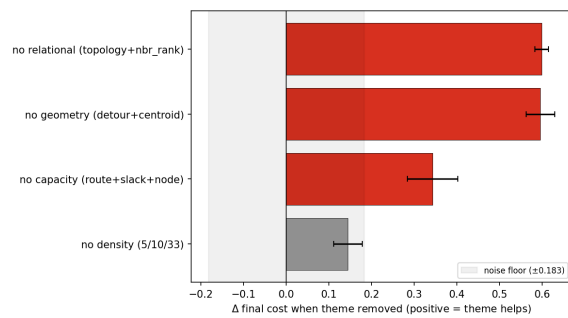


Figure 11: Thematic ablation (Table 9): removing the relational or geometry cluster costs ≈ 0.6 routing units each.

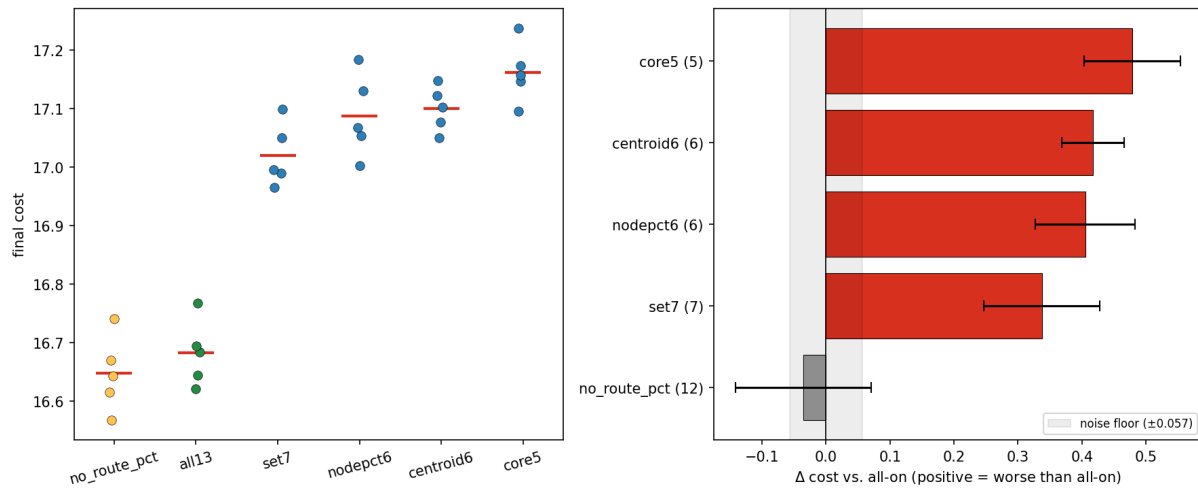


Figure 12: Best-set validation, per-seed view. Left: per-seed final cost — the 12-group and full-set clusters overlap and sit cleanly below every pruned set. Right: seed-paired Δ vs. the full set — the pruned sets (red) clear the noise floor by a wide margin; the 12-group set (green) sits below zero.

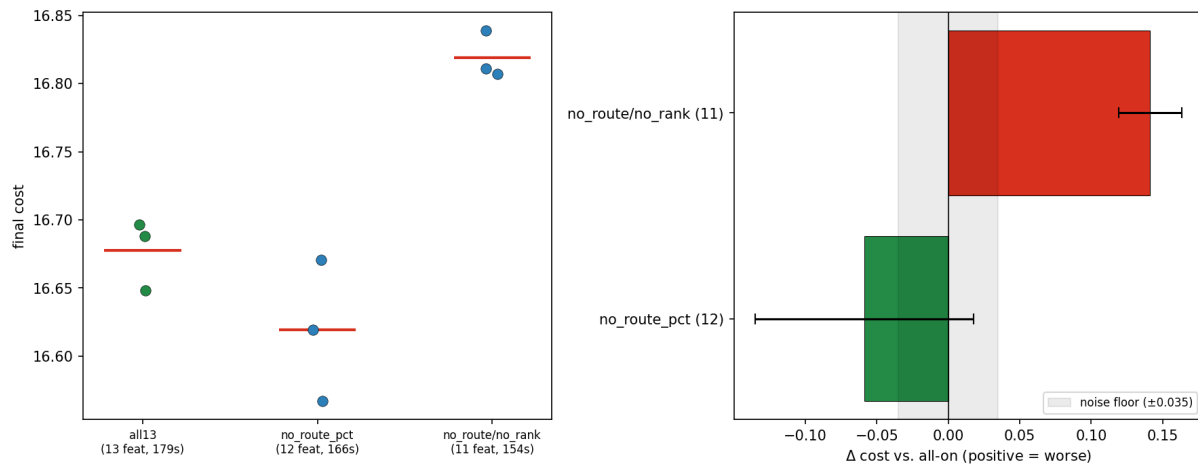


Figure 13: Targeted refinements (Table 10). Left: per-seed final cost — dropping the load ratio overlaps the full set, while additionally dropping the neighbor ranks lifts the whole cluster. Right: seed-paired Δ vs. the full set — the load-ratio removal (green) sits inside the noise floor, the neighbor-rank removal (red) clears it decisively.

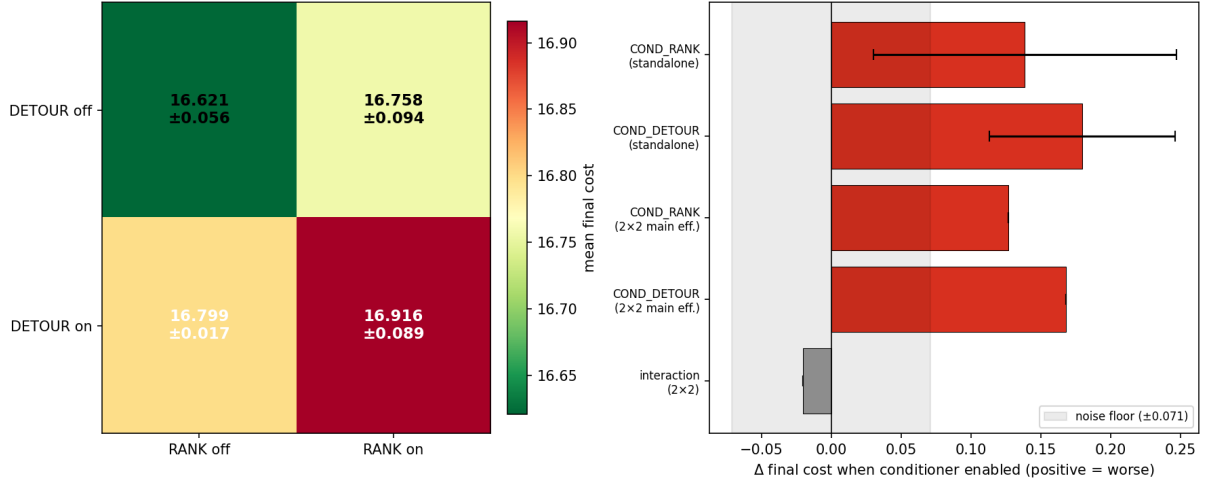


Figure 14: Conditional pair descriptors. Left: 2×2 set of configurations of mean final cost — the ‘both-off’ configuration (plain retained set) is best, ‘both-on’ is worst. Right: standalone deltas and 2×2 main effects — both descriptors sit clearly beyond the noise floor on the harmful side; the interaction is within it.

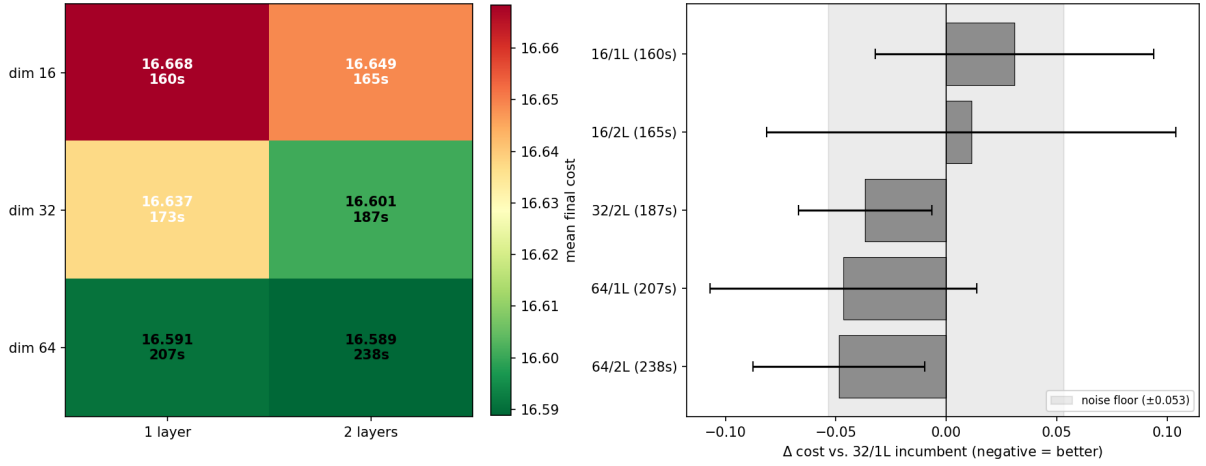


Figure 15: Actor architecture set of configurations (Table 12). Left: mean final cost and wall-clock per configuration (green = better). Right: seed-paired Δ against the 32×1 reference — every configuration falls inside the shaded noise band, so no architectural change is a significant win.

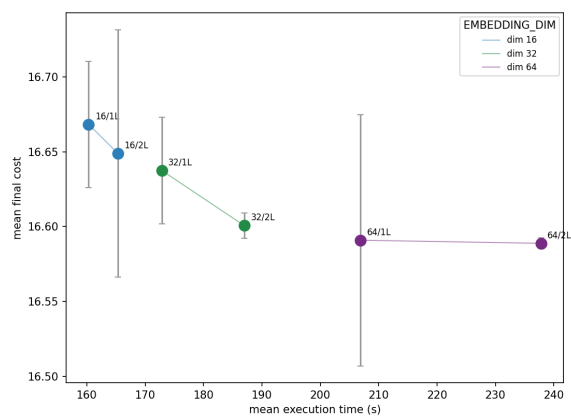


Figure 16: Cost against compute across the six architecture configurations, colored by embedding width. Quality improves only marginally along a steeply rising compute axis — the hallmark of a capacity plateau.

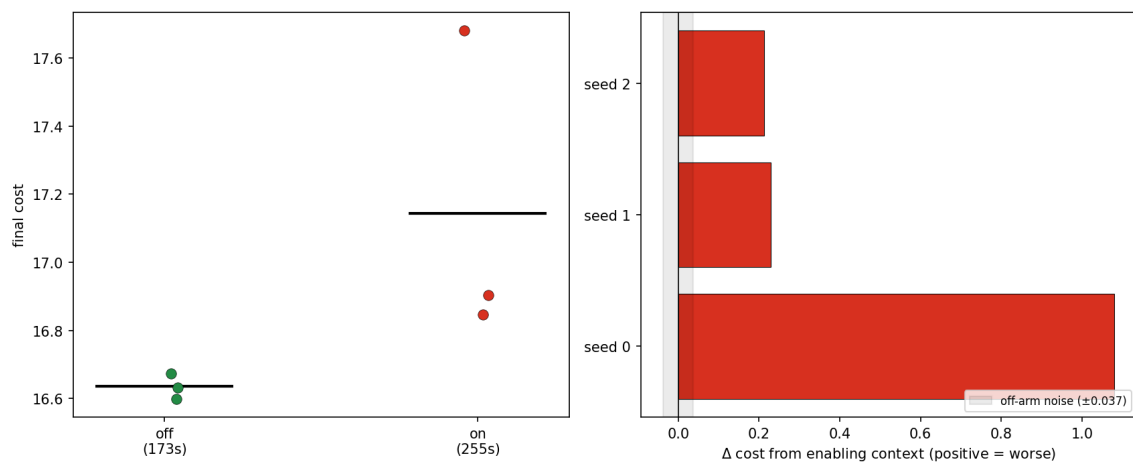


Figure 17: Global-context toggle. Left: per-seed final cost (black bars = means); the context-on arm is both worse and far more variable. Right: per-seed Δ from enabling context — every seed is positive (worse), one dramatically so, so the degradation is not a variance artefact.

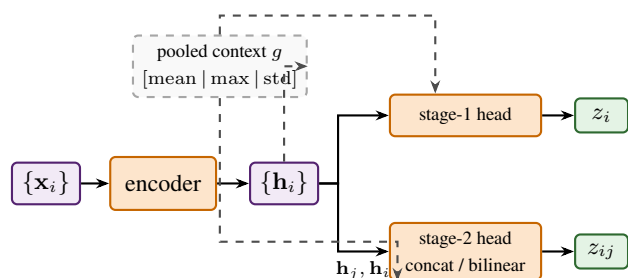


Figure 18: Shared-encoder actor variant. Each node is encoded once into an embedding h_i ; two lightweight heads then score the two stages over the shared embeddings, either by concatenation or as a bilinear compatibility between the selected node and each candidate. An optional pooled summary g of all node embeddings (dashed) is appended to both heads' inputs, giving every score access to instance-level context.

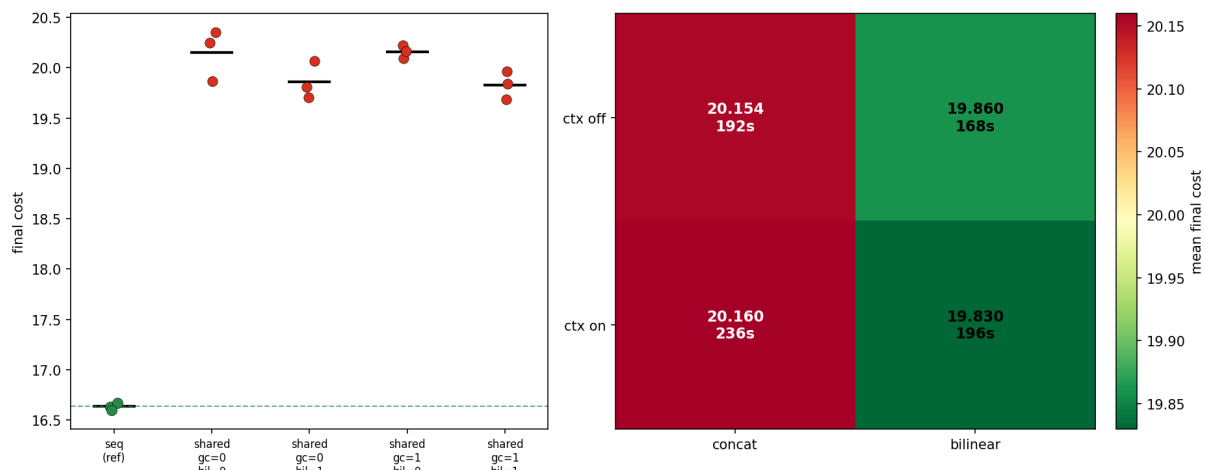


Figure 19: Shared encoder against independent scorers (Table 13). Left: per-seed final cost, reference actor (green) against the four shared configurations (red); the dashed line marks the reference mean, and every shared seed sits far above it. Right: the shared 2×2 set of configurations — the bilinear head beats concatenation and context is roughly neutral, but all four configurations are ≈ 3 units worse than the reference.

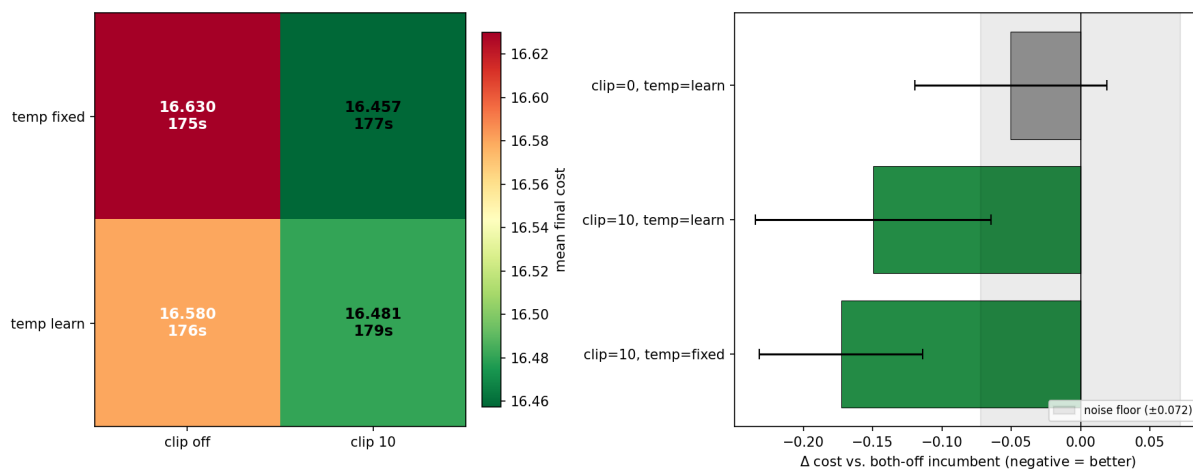


Figure 20: Distribution shaping (Table 14). Left: 2×2 set of configurations of mean final cost and time (green = better); the clipping-on, fixed-temperature configuration is the best. Right: seed-paired Δ against the both-off reference — only logit clipping escapes the noise band, and stacking the learnable temperature on top of it does not help.

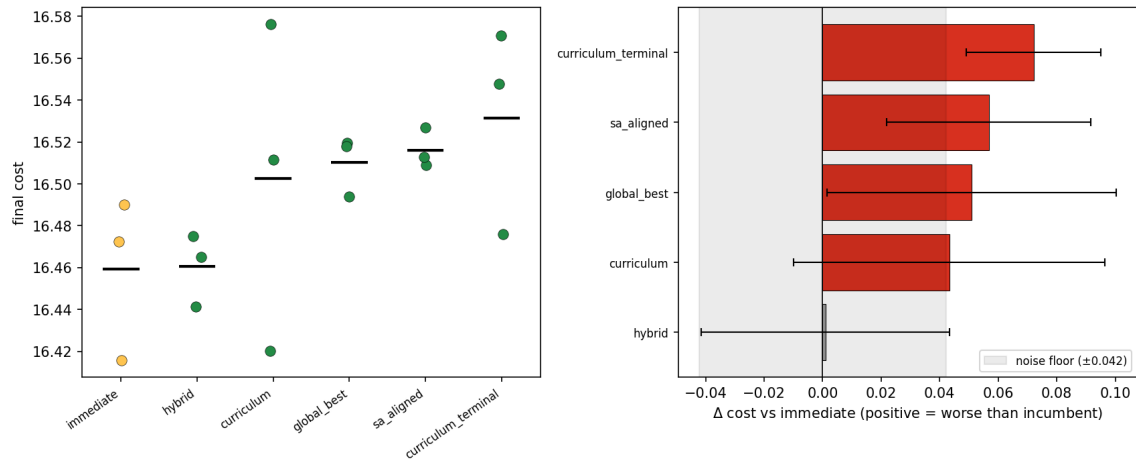


Figure 21: Zoom on the stable reward cluster. Left: per-seed final cost (black bars = means); the arms overlap heavily. Right: seed-paired Δ against the reference immediate reward — the fixed mixture sits inside the noise band (a tie), the remaining dense variants are marginally but consistently above zero.

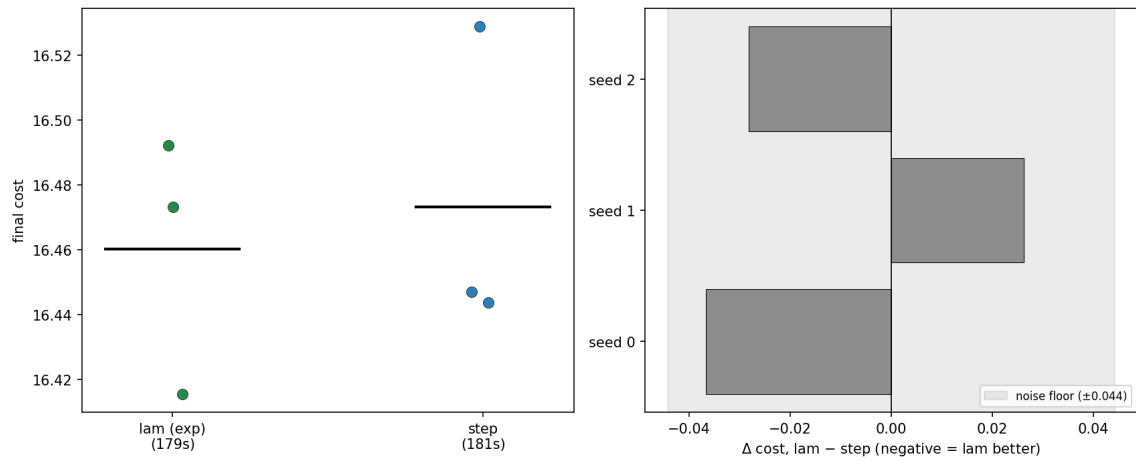


Figure 22: Cooling-schedule shape comparison. Left: per-seed final cost for the two schedules (black bars = means); the clusters overlap heavily. Right: per-seed paired Δ (geometric – step) — every bar lies inside the noise band and the sign is not consistent, the signature of a tie rather than an effect.

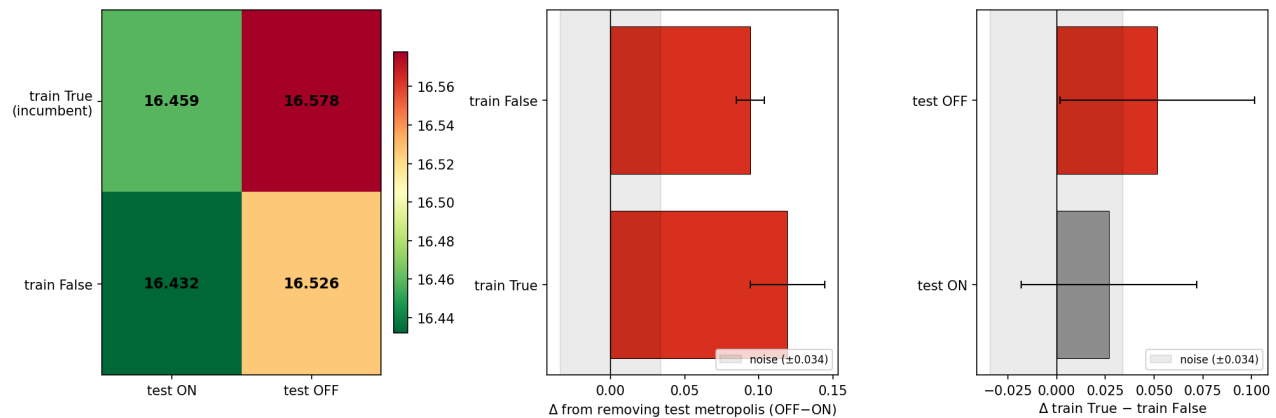


Figure 23: Metropolis train $\times$ test cross (Table 17). Left: 2×2 heatmap of mean final cost (greener = better). Center: cost of removing test-time Metropolis (off – on) at each training setting — both bars are well past the noise band, the dominant effect. Right: training-flag effect (on – off) at each inference — the test-on bar sits inside the band (a tie), test-off just clears it.

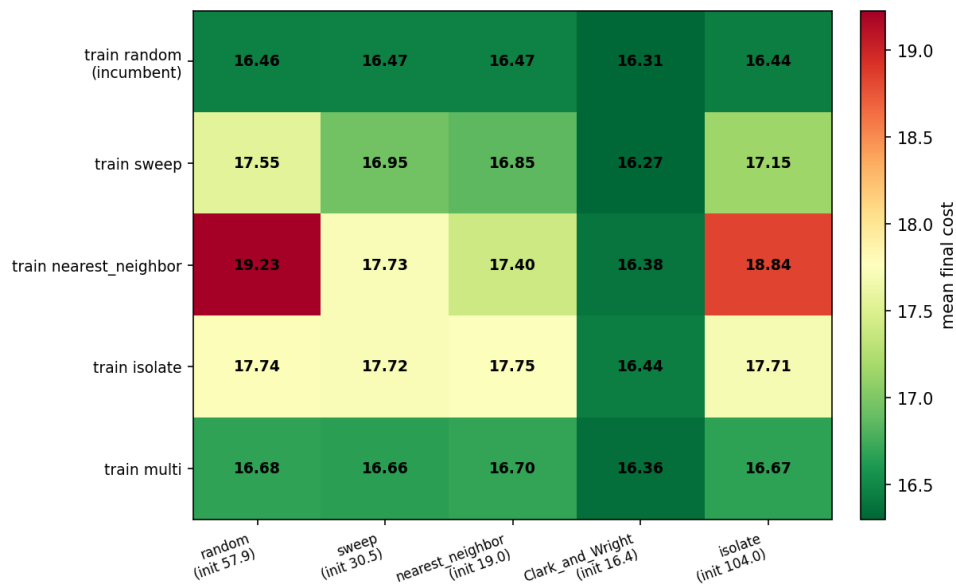


Figure 24: Initialization cross (Table 18): rows = training construction, columns = test-time construction, greener = lower final cost. The random-trained and mixed-trained rows are uniformly good; the three structured-training rows are poor and only recover in the Clarke–Wright column, where the construction itself already did the work.

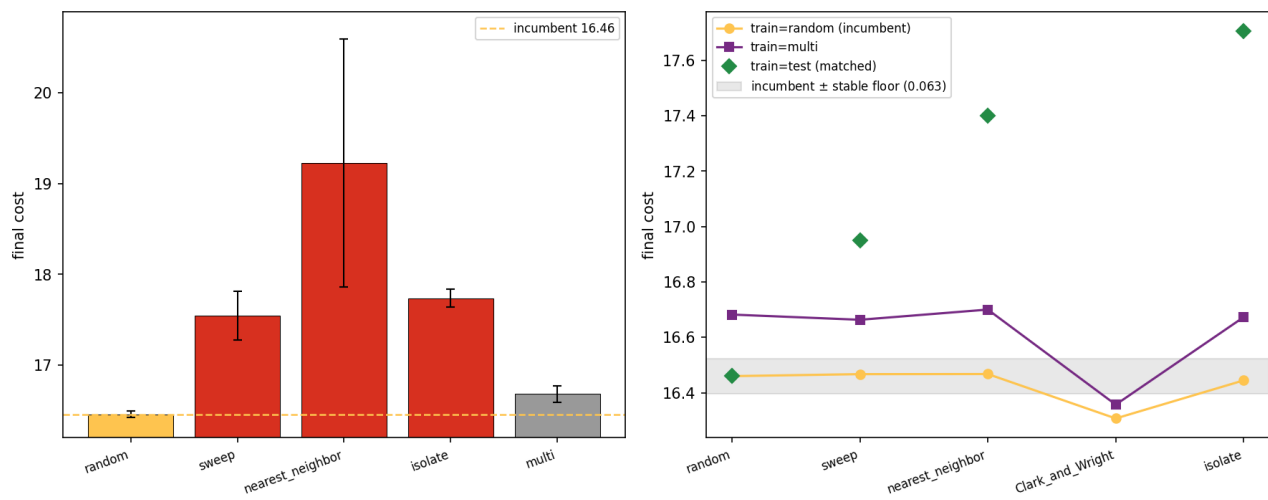


Figure 25: Left: final cost by training construction at the standard random-start inference — random (gold) is far below the rest, with mixed (gray) a close second. Right: final cost of selected actors across the five test constructions: the random-trained actor (gold) is nearly flat and mostly inside the stable-floor band, the mixed actor (purple) sits uniformly a touch above it, and the matched train/test diagonal (green diamonds) climbs away — ruling out a train/test-matching explanation for the training-construction effect.